Waymo_AV report

Waymo_AV — Robust Joint Motion Forecasting Under Missing and Noisy Observations

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0. Key results

Development set (500 scenes; checkpoint selection; NOT held-out) (per-method means over seeds; CIs and paired differences in the tables below)

method	clean joint minADE@6 (m)	core-suite joint minADE@6 (m)	clean overlap, own 6 outputs (%)	clean offroad (%)	n seeds
constant velocity	10.82	11.25	16.6	5.4	1
independent decoder	3.90 (seeds: 3.87, 4.02, 3.80)	7.87 (seeds: 8.10, 8.06, 7.46)	18.8 (seeds: 17.7, 18.4, 20.3)	5.8 (seeds: 6.0, 5.5, 5.9)	3
joint M0 (clean-trained)	3.74 (seeds: 3.78, 3.67, 3.77)	7.52 (seeds: 7.81, 7.31, 7.44)	15.0 (seeds: 14.6, 15.2, 15.4)	5.8 (seeds: 5.0, 5.8, 6.7)	3
joint M1 (corruption-trained)	3.51 (seeds: 3.56, 3.48, 3.50)	4.26 (seeds: 4.33, 4.23, 4.22)	14.9 (seeds: 15.8, 14.2, 14.7)	4.7 (seeds: 4.9, 4.7, 4.7)	3
M1 + best-6-of-32 reward selection	3.31 (seeds: 3.33, 3.27, 3.33)	4.06 (seeds: 4.09, 4.02, 4.08)	1.2 (seeds: 1.1, 0.6, 1.8)	1.6 (seeds: 1.6, 1.6, 1.6)	3
M1 + extra SFT	3.47 (seeds: 3.45, 3.47, 3.49)	4.25 (seeds: 4.30, 4.21, 4.25)	14.6 (seeds: 15.8, 13.6, 14.4)	4.8 (seeds: 5.0, 4.4, 4.9)	3

method	clean joint minADE@6 (m)	core-suite joint minADE@6 (m)	clean overlap, own 6 outputs (%)	clean offroad (%)	n seeds
M1 + reward-filtered SFT	3.55 (seeds: 3.61, 3.56, 3.49)	4.36 (seeds: 4.38, 4.39, 4.30)	14.4 (seeds: 15.2, 14.3, 13.7)	4.8 (seeds: 5.0, 4.6, 4.8)	3
M1 + bounded RL	4.35 (seeds: 4.50, 4.41, 4.14)	5.64 (seeds: 5.91, 5.75, 5.26)	5.3 (seeds: 5.3, 4.3, 6.2)	3.5 (seeds: 3.7, 3.3, 3.5)	3

- M1 vs M0 core-suite joint minADE@6 (seed of reference): -44.6 % [-47.3, -41.9] (group-level bootstrap). Per-seed verdicts against the preregistered thresholds: see the comparison section.
- FINAL held-out results: pending

RL outcome (D-031): recorded as mixed/negative under the frozen comparison rule — see section 3a (Q5/Q6) and DECISIONS.md.

1. Setup

- Dataset: { "dataset": "Waymo Open Motion Dataset", "version": "1.3.0", "raw_subdir": "womd_v1_3_0", "format": "scenario tfrecord", "bucket": "gs://waymo_open_dataset_motion_v_1_3_0", "listed": "2026-11-03", "listing_sha256": "d34a243222ae7bc2443da13cc8691847e2b34455f7dfb82a81798c091be3cfa1" }
- Calibration: { "n_scenes": 500, "tokenizer": { "saturation_rate": 0.0008159166509759618, "recon_err_mean_m": 0.09240365773439407, "recon_err_p99_m": 0.16335155069828033, "recon_err_max_m": 3.338541030883789, "vocab_used": 95, "top10_tokens": [[84, 3102], [71, 1884], [97, 1823], [85, 928], [83, 854], [58, 799], [110, 636], [70, 628], [96, 592], [72, 589]], "zero_token_frac": 0.19469026548672566 }, "d0_speed_mps": { "mean": 6.568544387817383, "p50": 5.209609031677246, "p99": 22.173171997070312, "max": 32.964210510253906 }, "history_valid_frac_targets": 0.9890802556818181, "future_valid_frac_targets": 0.9957275390625, "initial_overlap_rate_gt": 0.001953125, "future_overlap_rate_gt": 0.032421875, "transform_roundtrip_max_err_m": 5.684341886080802e-14, "d0_init": { "mean_err_m_vs_true_first_half_second": 0.24413703676310622, "p99_err_m": 0.84452919952413, "zero_velocity_err_m": 3.2363052368164062 }, "cv_baseline_clean": { "joint_minADE_8s": 10.822137399683946, "joint_minFDE_8s": 27.731932333955744 }, "target_types": { "1": 895, "2": 73, "3": 32 }, "offroad_oracle": { "sign_+1": 0.009799354392039928, "sign_-1": 0.99020064560796, "chosen_sign": 1,

"oracle_offroad_frac_with_chosen_sign": 0.009799354392039928,
"metric_valid": true } }

- Decisions: see DECISIONS.md; metric definitions: docs/metrics.md; preregistered thresholds: PLAN.md.

2. Runs

run	kind	status	step	params	best dev score
extra_sft_pilot_s0	train	completed	600	1787049	3.94908966326966
extra_sft_pilot_s1	train	completed	600	1787049	3.93531043102798
extra_sft_pilot_s2	train	completed	600	1787049	3.90064741090491
filtered_sft_pilot_s0	filtered_sft	completed	600		
filtered_sft_pilot_s1	filtered_sft	completed	600		
filtered_sft_pilot_s2	filtered_sft	completed	600		
m0_overfit	train	completed	600	1787049	0.09314770238688
m0_pilot_s0	train	completed	12000	1787049	3.82606548388915
m0_pilot_s1	train	completed	12000	1787049	3.74378330702582
m0_pilot_s2	train	completed	12000	1787049	3.59648358112898
m1_pilot_s0	train	completed	12000	1787049	3.98811603750914
m1_pilot_s1	train	completed	12000	1787049	3.84353757475528
m1_pilot_s2	train	completed	12000	1787049	3.88669393636379
mind_pilot_s0	train	completed	12000	1787049	3.84350423648226
mind_pilot_s1	train	completed	12000	1787049	4.05884199395848
mind_pilot_s2	train	completed	12000	1787049	3.83129117551994
rl_pilot_s0	rl	completed	600		
rl_pilot_s1	rl	completed	600		
rl_pilot_s2	rl	completed	600		

GPU-hours used (all runs): 2.27

2b. Metric qualification (before any post-training)

- Logged-future overlap audit v2 (D-028a; 500 dev scenes): reference rate on jointly valid intervals 1.0 % (5 scenes; classes {‘heading_inference’: 2, ‘annotation_contact’: 3}); the old unmasked full-horizon test flagged 15 scenes, 10 of them only on zero-filled invalid label steps. Superseded v1: manifests/overlap_floor_audit_v1_superseded.json.
- Generated-trajectory overlap inspection (D-028b; m0_pilot_s0, clean): 449/2994 (scene, sample) pairs flagged (15.0 %); verdicts {‘approximation_sensitive’: 56, ‘confirmed_geometric’: 324, ‘ambiguous’: 69}; by stratum {‘type:1-1’: {‘approximation_sensitive’: 40, ‘confirmed_geometric’: 252, ‘ambiguous’: 50}, ‘turning’: {‘approximation_sensitive’: 36, ‘ambiguous’: 26, ‘confirmed_geometric’:

134}, ‘straight_moving’: {‘confirmed_geometric’: 190, ‘ambiguous’: 32, ‘approximation_sensitive’: 20}, ‘stationary’: {‘ambiguous’: 13, ‘approximation_sensitive’: 2}, ‘type:1-2’: {‘approximation_sensitive’: 12, ‘ambiguous’: 17, ‘confirmed_geometric’: 48}, ‘type:1-3’: {‘approximation_sensitive’: 4, ‘confirmed_geometric’: 24, ‘ambiguous’: 2}}. The RL overlap reward is an approximate geometric-consistency objective, not a verified safety measure.

- Penetration depth does not separate logged-reference flags from model flags (p50 0.75 m vs 0.98 m); no depth margin is used.
- Offroad: sign convention +1 calibrated by log playback (manifests/metric_config.json); KD-tree nearest-segment implementation verified against brute force (tests).

3. Development-set results (checkpoint selection, sampling freeze)

Selection rules: M0/Mind checkpoints = lowest clean joint minADE@6 on the first 256 dev scenes; M1 = lowest mean of clean and core-suite joint minADE@6 on the same scenes (D-017). These selection scores are NOT comparable across models; every table below is a matched evaluation of the selected checkpoints on identical scenes, conditions and sampling.

method	condition	n	joint minADE@6 8s	joint minFDE@6 8s	marginal minADE 8s	h
cp	clean	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [(0
cv	clean	500	10.822 [10.236, 11.404]	27.732 [26.190, 29.256]	10.817 [10.236, 11.402]	0 [(0
extra_sft_pilot_s0	clean	500	3.454 [3.200, 3.706]	8.442 [7.750, 9.133]	2.968 [2.730, 3.198]	0 [(0
extra_sft_pilot_s1	clean	500	3.470 [3.241, 3.691]	8.401 [7.767, 9.049]	2.956 [2.747, 3.162]	0 [(0
extra_sft_pilot_s2	clean	500	3.493 [3.238, 3.732]	8.375 [7.672, 9.053]	2.997 [2.763, 3.228]	0 [(0
filtered_sft_pilot_s0	clean	500	3.609 [3.334, 3.877]	8.667 [7.953, 9.391]	3.064 [2.809, 3.312]	0 [(0
filtered_sft_pilot_s1	clean	500	3.559 [3.321, 3.803]	8.677 [8.000, 9.368]	3.081 [2.856, 3.305]	0 [(0
filtered_sft_pilot_s2	clean	500	3.488 [3.234,	8.411 [7.690,	2.984 [2.757,	0 [(

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	h
			3.728]	9.084]	3.208]	0
m0_pilot_s0	clean	500	3.776 [3.522, 4.037]	9.195 [8.468, 9.916]	3.185 [2.947, 3.413]	0 [0
m0_pilot_s1	clean	500	3.666 [3.417, 3.915]	8.672 [8.003, 9.338]	3.048 [2.816, 3.271]	0 [0
m0_pilot_s2	clean	500	3.765 [3.491, 4.027]	9.125 [8.428, 9.842]	3.157 [2.910, 3.397]	0 [0
m1_pilot_s0	clean	500	3.561 [3.310, 3.811]	8.418 [7.739, 9.089]	3.034 [2.806, 3.259]	0 [0
m1_pilot_s1	clean	500	3.481 [3.234, 3.724]	8.424 [7.734, 9.092]	2.993 [2.756, 3.223]	0 [0
m1_pilot_s2	clean	500	3.495 [3.246, 3.728]	8.360 [7.680, 9.019]	2.959 [2.733, 3.185]	0 [0
mind_pilot_s0	clean	500	3.871 [3.598, 4.137]	9.522 [8.764, 10.272]	3.270 [3.020, 3.528]	0 [0
mind_pilot_s1	clean	500	4.020 [3.755, 4.277]	9.724 [9.001, 10.440]	3.309 [3.074, 3.541]	0 [0
mind_pilot_s2	clean	500	3.796 [3.544, 4.053]	9.109 [8.437, 9.796]	3.147 [2.914, 3.380]	0 [0
oracle_log	clean	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	clean	500	4.504 [4.225, 4.755]	11.114 [10.349, 11.874]	3.984 [3.724, 4.223]	0 [0
rl_pilot_s1	clean	500	4.409 [4.147, 4.663]	10.856 [10.135, 11.545]	3.906 [3.646, 4.148]	0 [0
rl_pilot_s2	clean	500	4.139 [3.864, 4.404]	10.119 [9.368, 10.894]	3.603 [3.340, 3.855]	0 [0
select_m1_s0	clean	500	3.331 [3.081, 3.586]	7.818 [7.131, 8.509]	2.859 [2.622, 3.089]	0 [0
select_m1_s1	clean	500	3.273 [3.040,	7.899 [7.269,	2.823 [2.600,	0 [

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	h
			3.509]	8.549]	3.047]	0
select_m1_s2	clean	500	3.334 [3.102, 3.564]	8.073 [7.412, 8.728]	2.863 [2.640, 3.076]	0 [0
cp	compound_0	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [0
cv	compound_0	500	11.868 [11.308, 12.458]	29.125 [27.641, 30.505]	11.861 [11.302, 12.454]	0 [0
extra_sft_pilot_s0	compound_0	500	5.347 [5.079, 5.613]	11.493 [10.758, 12.229]	4.602 [4.350, 4.859]	0 [0
extra_sft_pilot_s1	compound_0	500	5.193 [4.924, 5.469]	11.230 [10.493, 11.992]	4.440 [4.182, 4.708]	0 [0
extra_sft_pilot_s2	compound_0	500	5.321 [5.041, 5.600]	11.375 [10.655, 12.120]	4.487 [4.218, 4.758]	0 [0
filtered_sft_pilot_s0	compound_0	500	5.416 [5.131, 5.705]	11.363 [10.652, 12.112]	4.657 [4.391, 4.932]	0 [0
filtered_sft_pilot_s1	compound_0	500	5.623 [5.319, 5.928]	12.107 [11.329, 12.901]	4.842 [4.550, 5.131]	0 [0
filtered_sft_pilot_s2	compound_0	500	5.439 [5.141, 5.740]	11.959 [11.182, 12.755]	4.589 [4.321, 4.865]	0 [0
m0_pilot_s0	compound_0	500	10.052 [9.579, 10.533]	21.337 [20.169, 22.545]	8.657 [8.219, 9.097]	0 [0
m0_pilot_s1	compound_0	500	10.404 [9.896, 10.925]	22.592 [21.395, 23.819]	9.053 [8.583, 9.548]	0 [0
m0_pilot_s2	compound_0	500	10.272 [9.791, 10.745]	22.026 [20.900, 23.332]	9.086 [8.626, 9.541]	0 [0
m1_pilot_s0	compound_0	500	5.334 [5.067, 5.619]	11.194 [10.445, 11.961]	4.546 [4.301, 4.800]	0 [0
m1_pilot_s1	compound_0	500	5.264 [4.998, 5.542]	11.307 [10.592, 12.053]	4.495 [4.231, 4.762]	0 [0
m1_pilot_s2	compound_0	500	5.265 [4.987,	11.351 [10.625,	4.443 [4.190,	0 [

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	c r h
			5.559]	12.091]	4.708]	0
mind_pilot_s0	compound_0	500	10.355 [9.888, 10.814]	21.527 [20.460, 22.659]	9.177 [8.737, 9.599]	0 [0
mind_pilot_s1	compound_0	500	11.323 [10.793, 11.823]	24.499 [23.199, 25.774]	9.707 [9.221, 10.170]	0 [0
mind_pilot_s2	compound_0	500	9.809 [9.316, 10.296]	20.813 [19.687, 21.951]	8.628 [8.164, 9.079]	0 [0
oracle_log	compound_0	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	compound_0	500	7.717 [7.360, 8.088]	16.986 [16.048, 17.941]	6.986 [6.640, 7.355]	0 [0
rl_pilot_s1	compound_0	500	7.494 [7.123, 7.865]	16.806 [15.807, 17.810]	6.743 [6.380, 7.114]	0 [0
rl_pilot_s2	compound_0	500	6.767 [6.397, 7.126]	15.266 [14.333, 16.170]	5.965 [5.604, 6.308]	0 [0
select_m1_s0	compound_0	500	5.002 [4.738, 5.285]	10.655 [9.948, 11.382]	4.350 [4.103, 4.613]	0 [0
select_m1_s1	compound_0	500	5.114 [4.809, 5.419]	10.956 [10.172, 11.744]	4.412 [4.122, 4.710]	0 [0
select_m1_s2	compound_0	500	5.107 [4.817, 5.404]	11.034 [10.282, 11.871]	4.425 [4.137, 4.712]	0 [0
cp	missing_0.1	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [0
cv	missing_0.1	500	10.814 [10.238, 11.395]	27.719 [26.177, 29.231]	10.808 [10.219, 11.387]	0 [0
extra_sft_pilot_s0	missing_0.1	500	3.440 [3.198, 3.691]	8.281 [7.576, 8.979]	2.869 [2.652, 3.090]	0 [0
extra_sft_pilot_s1	missing_0.1	500	3.511 [3.275, 3.754]	8.345 [7.706, 9.014]	2.931 [2.713, 3.141]	0 [0
extra_sft_pilot_s2	missing_0.1	500	3.612 [3.356,	8.647 [7.916,	3.068 [2.822,	0 [

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	h
			3.861]	9.363]	3.287]	0
filtered_sft_pilot_s0	missing_0.1	500	3.524 [3.262, 3.769]	8.429 [7.709, 9.093]	3.018 [2.780, 3.250]	0 [0
filtered_sft_pilot_s1	missing_0.1	500	3.590 [3.332, 3.833]	8.640 [7.959, 9.325]	3.091 [2.849, 3.332]	0 [0
filtered_sft_pilot_s2	missing_0.1	500	3.545 [3.297, 3.795]	8.582 [7.872, 9.284]	3.056 [2.813, 3.290]	0 [0
m0_pilot_s0	missing_0.1	500	3.941 [3.691, 4.195]	9.485 [8.754, 10.208]	3.348 [3.109, 3.581]	0 [0
m0_pilot_s1	missing_0.1	500	3.785 [3.536, 4.043]	9.044 [8.365, 9.746]	3.209 [2.981, 3.447]	0 [0
m0_pilot_s2	missing_0.1	500	3.949 [3.677, 4.211]	9.587 [8.829, 10.349]	3.393 [3.141, 3.639]	0 [0
m1_pilot_s0	missing_0.1	500	3.462 [3.213, 3.719]	8.382 [7.674, 9.084]	2.919 [2.687, 3.159]	0 [0
m1_pilot_s1	missing_0.1	500	3.533 [3.271, 3.784]	8.575 [7.902, 9.282]	2.966 [2.725, 3.201]	0 [0
m1_pilot_s2	missing_0.1	500	3.463 [3.197, 3.698]	8.386 [7.677, 9.062]	2.983 [2.737, 3.199]	0 [0
mind_pilot_s0	missing_0.1	500	4.039 [3.746, 4.314]	9.863 [9.058, 10.636]	3.395 [3.124, 3.653]	0 [0
mind_pilot_s1	missing_0.1	500	4.076 [3.808, 4.335]	9.719 [9.004, 10.430]	3.370 [3.140, 3.589]	0 [0
mind_pilot_s2	missing_0.1	500	4.173 [3.896, 4.467]	9.928 [9.160, 10.704]	3.505 [3.245, 3.770]	0 [0
oracle_log	missing_0.1	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	missing_0.1	500	4.485 [4.205, 4.760]	11.078 [10.296, 11.857]	3.994 [3.724, 4.250]	0 [0
rl_pilot_s1	missing_0.1	500	4.428 [4.163,	10.935 [10.181,	3.861 [3.603,	0 [

method	condition	n	joint minADE@6 8s	joint minFDE@6 8s	marginal minADE 8s	h
			4.685]	11.660]	4.106]	0
rl_pilot_s2	missing_0.1	500	4.197 [3.916, 4.464]	10.380 [9.593, 11.080]	3.652 [3.386, 3.900]	0 [0
select_m1_s0	missing_0.1	500	3.384 [3.134, 3.637]	8.064 [7.343, 8.802]	2.913 [2.681, 3.160]	0 [0
select_m1_s1	missing_0.1	500	3.290 [3.041, 3.531]	7.852 [7.243, 8.498]	2.843 [2.620, 3.068]	0 [0
select_m1_s2	missing_0.1	500	3.397 [3.136, 3.630]	8.081 [7.381, 8.767]	2.914 [2.669, 3.137]	0 [0
cp	missing_0.3	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [0
cv	missing_0.3	500	10.768 [10.193, 11.348]	27.647 [26.120, 29.150]	10.763 [10.177, 11.347]	0 [0
extra_sft_pilot_s0	missing_0.3	500	3.544 [3.301, 3.788]	8.470 [7.786, 9.129]	3.022 [2.784, 3.248]	0 [0
extra_sft_pilot_s1	missing_0.3	500	3.512 [3.270, 3.756]	8.423 [7.794, 9.088]	2.991 [2.766, 3.220]	0 [0
extra_sft_pilot_s2	missing_0.3	500	3.475 [3.236, 3.713]	8.386 [7.714, 9.043]	2.971 [2.746, 3.196]	0 [0
filtered_sft_pilot_s0	missing_0.3	500	3.528 [3.281, 3.776]	8.394 [7.710, 9.076]	3.014 [2.778, 3.240]	0 [0
filtered_sft_pilot_s1	missing_0.3	500	3.584 [3.339, 3.825]	8.520 [7.853, 9.163]	3.028 [2.803, 3.248]	0 [0
filtered_sft_pilot_s2	missing_0.3	500	3.595 [3.343, 3.838]	8.715 [8.030, 9.436]	3.052 [2.821, 3.274]	0 [0
m0_pilot_s0	missing_0.3	500	4.309 [4.041, 4.591]	10.347 [9.582, 11.123]	3.628 [3.377, 3.882]	0 [0
m0_pilot_s1	missing_0.3	500	4.348 [4.042, 4.640]	10.162 [9.364, 10.958]	3.717 [3.440, 3.991]	0 [0
m0_pilot_s2	missing_0.3	500	4.110 [3.829,	9.837 [9.117,	3.496 [3.238,	0 [

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE8s	h
			4.389]	10.592]	3.758]	0
m1_pilot_s0	missing_0.3	500	3.594 [3.330, 3.857]	8.441 [7.763, 9.125]	3.047 [2.798, 3.285]	0 [0
m1_pilot_s1	missing_0.3	500	3.502 [3.255, 3.745]	8.491 [7.840, 9.158]	2.999 [2.766, 3.234]	0 [0
m1_pilot_s2	missing_0.3	500	3.523 [3.278, 3.759]	8.365 [7.687, 9.017]	2.976 [2.747, 3.200]	0 [0
mind_pilot_s0	missing_0.3	500	4.434 [4.119, 4.735]	10.728 [9.914, 11.532]	3.805 [3.517, 4.078]	0 [0
mind_pilot_s1	missing_0.3	500	4.788 [4.488, 5.092]	11.472 [10.617, 12.287]	3.976 [3.710, 4.253]	0 [0
mind_pilot_s2	missing_0.3	500	4.532 [4.247, 4.833]	10.679 [9.913, 11.453]	3.749 [3.486, 4.019]	0 [0
oracle_log	missing_0.3	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	missing_0.3	500	4.491 [4.215, 4.749]	11.010 [10.233, 11.778]	3.999 [3.722, 4.245]	0 [0
rl_pilot_s1	missing_0.3	500	4.419 [4.143, 4.690]	10.884 [10.155, 11.612]	3.843 [3.586, 4.093]	0 [0
rl_pilot_s2	missing_0.3	500	4.223 [3.943, 4.496]	10.327 [9.540, 11.119]	3.662 [3.393, 3.909]	0 [0
select_m1_s0	missing_0.3	500	3.448 [3.182, 3.709]	8.348 [7.584, 9.120]	2.953 [2.709, 3.194]	0 [0
select_m1_s1	missing_0.3	500	3.426 [3.175, 3.664]	8.159 [7.527, 8.821]	2.974 [2.732, 3.202]	0 [0
select_m1_s2	missing_0.3	500	3.551 [3.291, 3.803]	8.548 [7.828, 9.285]	3.075 [2.827, 3.315]	0 [0
cp	missing_0.5	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE8s	c 1 h
cv	missing_0.5	500	10.762 [10.179, 11.342]	27.588 [26.084, 29.067]	10.758 [10.173, 11.333]	0 [(0
extra_sft_pilot_s0	missing_0.5	500	3.637 [3.383, 3.908]	8.746 [8.045, 9.471]	3.115 [2.877, 3.371]	0 [(0
extra_sft_pilot_s1	missing_0.5	500	3.612 [3.368, 3.853]	8.625 [7.984, 9.307]	3.084 [2.846, 3.315]	0 [(0
extra_sft_pilot_s2	missing_0.5	500	3.792 [3.526, 4.047]	9.302 [8.565, 10.032]	3.267 [3.025, 3.518]	0 [(0
filtered_sft_pilot_s0	missing_0.5	500	3.754 [3.506, 3.996]	8.643 [7.985, 9.312]	3.210 [2.987, 3.433]	0 [(0
filtered_sft_pilot_s1	missing_0.5	500	3.696 [3.455, 3.958]	8.882 [8.199, 9.592]	3.179 [2.942, 3.421]	0 [(0
filtered_sft_pilot_s2	missing_0.5	500	3.784 [3.513, 4.053]	9.080 [8.344, 9.817]	3.246 [2.992, 3.499]	0 [(0
m0_pilot_s0	missing_0.5	500	4.776 [4.461, 5.097]	11.535 [10.622, 12.433]	4.090 [3.789, 4.396]	0 [(0
m0_pilot_s1	missing_0.5	500	4.937 [4.615, 5.253]	11.375 [10.561, 12.219]	4.175 [3.878, 4.473]	0 [(0
m0_pilot_s2	missing_0.5	500	4.630 [4.336, 4.910]	10.898 [10.135, 11.717]	3.926 [3.648, 4.193]	0 [(0
m1_pilot_s0	missing_0.5	500	3.658 [3.395, 3.924]	8.805 [8.080, 9.518]	3.132 [2.887, 3.376]	0 [(0
m1_pilot_s1	missing_0.5	500	3.612 [3.363, 3.865]	8.566 [7.904, 9.232]	3.044 [2.822, 3.275]	0 [(0
m1_pilot_s2	missing_0.5	500	3.762 [3.486, 4.017]	9.002 [8.221, 9.752]	3.219 [2.954, 3.468]	0 [(0
mind_pilot_s0	missing_0.5	500	5.148 [4.826, 5.461]	12.264 [11.403, 13.098]	4.302 [4.007, 4.586]	0 [(0
mind_pilot_s1	missing_0.5	500	5.464 [5.114, 5.815]	13.065 [12.148, 14.035]	4.541 [4.230, 4.854]	0 [(0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	h
mind_pilot_s2	missing_0.5	500	4.921 [4.617, 5.234]	11.471 [10.713, 12.326]	4.086 [3.808, 4.376]	0 [0
oracle_log	missing_0.5	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	missing_0.5	500	4.975 [4.671, 5.292]	12.220 [11.365, 13.055]	4.470 [4.176, 4.771]	0 [0
rl_pilot_s1	missing_0.5	500	4.766 [4.475, 5.053]	11.766 [10.964, 12.540]	4.201 [3.922, 4.475]	0 [0
rl_pilot_s2	missing_0.5	500	4.379 [4.087, 4.660]	10.681 [9.926, 11.445]	3.847 [3.575, 4.110]	0 [0
select_m1_s0	missing_0.5	500	3.566 [3.293, 3.832]	8.597 [7.853, 9.355]	3.066 [2.809, 3.320]	0 [0
select_m1_s1	missing_0.5	500	3.510 [3.261, 3.756]	8.253 [7.596, 8.949]	3.001 [2.773, 3.232]	0 [0
select_m1_s2	missing_0.5	500	3.570 [3.304, 3.838]	8.535 [7.833, 9.231]	3.082 [2.830, 3.337]	0 [0
cp	missing_0.7	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [0
cv	missing_0.7	500	11.427 [10.781, 12.093]	28.673 [27.113, 30.278]	11.427 [10.775, 12.090]	0 [0
extra_sft_pilot_s0	missing_0.7	500	4.244 [3.942, 4.537]	9.661 [8.965, 10.341]	3.604 [3.324, 3.885]	0 [0
extra_sft_pilot_s1	missing_0.7	500	4.151 [3.865, 4.455]	9.334 [8.640, 10.056]	3.577 [3.299, 3.866]	0 [0
extra_sft_pilot_s2	missing_0.7	500	4.163 [3.873, 4.477]	9.571 [8.833, 10.330]	3.570 [3.303, 3.857]	0 [0
filtered_sft_pilot_s0	missing_0.7	500	4.500 [4.189, 4.819]	10.070 [9.313, 10.863]	3.808 [3.532, 4.093]	0 [0
filtered_sft_pilot_s1	missing_0.7	500	4.445 [4.122, 4.792]	10.203 [9.418, 11.007]	3.830 [3.526, 4.150]	0 [0

method	condition	n	joint minADE@6 8s	joint minFDE@6 8s	marginal minADE 8s	h
filtered_sft_pilot_s2	missing_0.7	500	4.413 [4.074, 4.777]	10.063 [9.194, 10.875]	3.838 [3.505, 4.181]	0 [0
m0_pilot_s0	missing_0.7	500	5.819 [5.424, 6.233]	13.362 [12.418, 14.315]	4.951 [4.592, 5.330]	0 [0
m0_pilot_s1	missing_0.7	500	6.211 [5.800, 6.658]	14.030 [12.995, 15.126]	5.260 [4.905, 5.657]	0 [0
m0_pilot_s2	missing_0.7	500	5.835 [5.475, 6.207]	13.334 [12.454, 14.311]	4.956 [4.626, 5.303]	0 [0
m1_pilot_s0	missing_0.7	500	4.208 [3.911, 4.508]	9.571 [8.898, 10.268]	3.568 [3.292, 3.856]	0 [0
m1_pilot_s1	missing_0.7	500	4.219 [3.907, 4.561]	9.557 [8.790, 10.357]	3.635 [3.347, 3.946]	0 [0
m1_pilot_s2	missing_0.7	500	4.194 [3.912, 4.483]	9.599 [8.896, 10.356]	3.610 [3.341, 3.883]	0 [0
mind_pilot_s0	missing_0.7	500	6.139 [5.746, 6.537]	14.106 [13.208, 15.063]	5.232 [4.877, 5.599]	0 [0
mind_pilot_s1	missing_0.7	500	7.069 [6.590, 7.576]	16.427 [15.306, 17.596]	5.979 [5.549, 6.464]	0 [0
mind_pilot_s2	missing_0.7	500	6.309 [5.916, 6.730]	14.253 [13.265, 15.291]	5.461 [5.087, 5.858]	0 [0
oracle_log	missing_0.7	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	missing_0.7	500	5.527 [5.148, 5.875]	13.140 [12.280, 14.003]	5.022 [4.657, 5.354]	0 [0
rl_pilot_s1	missing_0.7	500	5.653 [5.232, 6.083]	12.951 [12.034, 13.906]	5.075 [4.660, 5.499]	0 [0
rl_pilot_s2	missing_0.7	500	5.051 [4.689, 5.418]	11.716 [10.820, 12.628]	4.444 [4.106, 4.800]	0 [0
select_m1_s0	missing_0.7	500	3.989 [3.708, 4.296]	8.946 [8.246, 9.657]	3.456 [3.176, 3.745]	0 [0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	c 1 h
select_m1_s1	missing_0.7	500	4.132 [3.825, 4.456]	9.385 [8.634, 10.169]	3.571 [3.291, 3.876]	0 [(0
select_m1_s2	missing_0.7	500	3.941 [3.654, 4.242]	9.047 [8.302, 9.800]	3.409 [3.136, 3.688]	0 [(0
cp	noise_0.1	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [(0
cv	noise_0.1	500	10.941 [10.362, 11.522]	27.861 [26.335, 29.377]	10.935 [10.352, 11.515]	0 [(0
extra_sft_pilot_s0	noise_0.1	500	4.118 [3.844, 4.396]	9.431 [8.671, 10.129]	3.502 [3.256, 3.758]	0 [(0
extra_sft_pilot_s1	noise_0.1	500	4.007 [3.760, 4.266]	9.185 [8.494, 9.868]	3.400 [3.167, 3.644]	0 [(0
extra_sft_pilot_s2	noise_0.1	500	4.123 [3.851, 4.393]	9.524 [8.782, 10.248]	3.516 [3.258, 3.771]	0 [(0
filtered_sft_pilot_s0	noise_0.1	500	4.084 [3.811, 4.345]	9.388 [8.641, 10.065]	3.471 [3.224, 3.717]	0 [(0
filtered_sft_pilot_s1	noise_0.1	500	4.115 [3.850, 4.381]	9.584 [8.895, 10.297]	3.519 [3.268, 3.770]	0 [(0
filtered_sft_pilot_s2	noise_0.1	500	4.118 [3.836, 4.402]	9.468 [8.734, 10.208]	3.509 [3.243, 3.772]	0 [(0
m0_pilot_s0	noise_0.1	500	6.226 [5.849, 6.606]	13.746 [12.764, 14.732]	5.251 [4.926, 5.585]	0 [(0
m0_pilot_s1	noise_0.1	500	6.066 [5.726, 6.403]	13.583 [12.724, 14.469]	5.276 [4.957, 5.594]	0 [(0
m0_pilot_s2	noise_0.1	500	5.710 [5.387, 6.042]	12.832 [12.013, 13.694]	4.903 [4.599, 5.204]	0 [(0
m1_pilot_s0	noise_0.1	500	4.205 [3.936, 4.487]	9.662 [8.926, 10.362]	3.467 [3.222, 3.719]	0 [(0
m1_pilot_s1	noise_0.1	500	3.943 [3.689, 4.198]	9.150 [8.501, 9.843]	3.361 [3.119, 3.605]	0 [(0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	h
m1_pilot_s2	noise_0.1	500	4.033 [3.766, 4.299]	9.280 [8.595, 10.011]	3.459 [3.214, 3.699]	0 [0
mind_pilot_s0	noise_0.1	500	6.600 [6.200, 7.010]	14.539 [13.573, 15.601]	5.683 [5.310, 6.058]	0 [0
mind_pilot_s1	noise_0.1	500	6.227 [5.840, 6.603]	13.702 [12.758, 14.678]	5.300 [4.951, 5.642]	0 [0
mind_pilot_s2	noise_0.1	500	5.535 [5.186, 5.860]	12.570 [11.678, 13.483]	4.740 [4.423, 5.033]	0 [0
oracle_log	noise_0.1	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	noise_0.1	500	5.621 [5.301, 5.932]	13.065 [12.232, 13.895]	4.943 [4.631, 5.247]	0 [0
rl_pilot_s1	noise_0.1	500	5.602 [5.280, 5.914]	13.277 [12.464, 14.057]	4.989 [4.675, 5.291]	0 [0
rl_pilot_s2	noise_0.1	500	5.050 [4.754, 5.345]	11.819 [11.038, 12.625]	4.391 [4.107, 4.685]	0 [0
select_m1_s0	noise_0.1	500	3.851 [3.588, 4.108]	8.795 [8.115, 9.473]	3.297 [3.056, 3.542]	0 [0
select_m1_s1	noise_0.1	500	3.892 [3.624, 4.157]	8.999 [8.274, 9.741]	3.352 [3.094, 3.601]	0 [0
select_m1_s2	noise_0.1	500	4.084 [3.771, 4.404]	9.603 [8.776, 10.450]	3.482 [3.193, 3.777]	0 [0
cp	noise_0.3	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [0
cv	noise_0.3	500	11.631 [11.098, 12.187]	28.848 [27.384, 30.278]	11.628 [11.094, 12.186]	0 [0
extra_sft_pilot_s0	noise_0.3	500	4.949 [4.674, 5.230]	10.786 [10.014, 11.545]	4.125 [3.876, 4.385]	0 [0
extra_sft_pilot_s1	noise_0.3	500	4.751 [4.499, 5.000]	10.432 [9.733, 11.118]	3.966 [3.736, 4.200]	0 [0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	h
extra_sft_pilot_s2	noise_0.3	500	4.862 [4.578, 5.146]	10.665 [9.896, 11.420]	4.036 [3.776, 4.291]	0 [0
filtered_sft_pilot_s0	noise_0.3	500	5.105 [4.831, 5.379]	11.073 [10.324, 11.804]	4.322 [4.066, 4.580]	0 [0
filtered_sft_pilot_s1	noise_0.3	500	5.100 [4.821, 5.377]	11.142 [10.427, 11.914]	4.303 [4.049, 4.561]	0 [0
filtered_sft_pilot_s2	noise_0.3	500	5.006 [4.732, 5.276]	11.105 [10.325, 11.861]	4.128 [3.871, 4.380]	0 [0
m0_pilot_s0	noise_0.3	500	12.417 [11.855, 12.942]	26.141 [24.902, 27.420]	10.425 [9.911, 10.894]	0 [0
m0_pilot_s1	noise_0.3	500	11.094 [10.567, 11.611]	23.840 [22.570, 25.145]	9.662 [9.173, 10.146]	0 [0
m0_pilot_s2	noise_0.3	500	11.428 [10.915, 11.953]	25.190 [24.029, 26.478]	10.107 [9.611, 10.603]	0 [0
m1_pilot_s0	noise_0.3	500	4.940 [4.678, 5.205]	10.918 [10.182, 11.646]	4.115 [3.863, 4.374]	0 [0
m1_pilot_s1	noise_0.3	500	4.857 [4.604, 5.112]	10.455 [9.789, 11.107]	4.007 [3.769, 4.257]	0 [0
m1_pilot_s2	noise_0.3	500	4.831 [4.560, 5.105]	10.721 [9.997, 11.459]	3.982 [3.745, 4.231]	0 [0
mind_pilot_s0	noise_0.3	500	12.800 [12.262, 13.310]	27.253 [26.041, 28.581]	10.941 [10.440, 11.440]	0 [0
mind_pilot_s1	noise_0.3	500	12.466 [11.906, 13.021]	27.252 [25.852, 28.592]	10.796 [10.274, 11.314]	0 [0
mind_pilot_s2	noise_0.3	500	11.034 [10.530, 11.566]	23.635 [22.321, 24.990]	9.700 [9.205, 10.213]	0 [0
oracle_log	noise_0.3	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	noise_0.3	500	7.223 [6.900, 7.560]	16.086 [15.206, 16.998]	6.425 [6.104, 6.755]	0 [0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE8s	c 1 h
rl_pilot_s1	noise_0.3	500	7.005 [6.663, 7.357]	15.631 [14.726, 16.539]	6.275 [5.930, 6.608]	0 [(0
rl_pilot_s2	noise_0.3	500	6.299 [5.969, 6.629]	14.259 [13.402, 15.122]	5.590 [5.269, 5.904]	0 [(0
select_m1_s0	noise_0.3	500	4.675 [4.397, 4.960]	10.369 [9.614, 11.117]	3.953 [3.701, 4.211]	0 [(0
select_m1_s1	noise_0.3	500	4.612 [4.331, 4.890]	10.047 [9.317, 10.759]	3.830 [3.569, 4.093]	0 [(0
select_m1_s2	noise_0.3	500	4.558 [4.283, 4.831]	10.074 [9.342, 10.818]	3.831 [3.579, 4.083]	0 [(0
cp	noise_0.5	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [(0
cv	noise_0.5	500	12.586 [12.052, 13.128]	30.255 [28.800, 31.669]	12.579 [12.045, 13.122]	0 [(0
extra_sft_pilot_s0	noise_0.5	500	6.123 [5.824, 6.409]	12.973 [12.227, 13.732]	5.078 [4.803, 5.342]	0 [(0
extra_sft_pilot_s1	noise_0.5	500	5.847 [5.571, 6.141]	12.508 [11.822, 13.227]	4.892 [4.621, 5.159]	0 [(0
extra_sft_pilot_s2	noise_0.5	500	5.636 [5.379, 5.900]	11.975 [11.271, 12.641]	4.619 [4.382, 4.865]	0 [(0
filtered_sft_pilot_s0	noise_0.5	500	6.278 [5.972, 6.576]	13.124 [12.364, 13.906]	5.229 [4.932, 5.515]	0 [(0
filtered_sft_pilot_s1	noise_0.5	500	6.251 [5.962, 6.546]	13.210 [12.461, 13.964]	5.204 [4.939, 5.487]	0 [(0
filtered_sft_pilot_s2	noise_0.5	500	5.770 [5.481, 6.074]	12.238 [11.521, 12.953]	4.819 [4.558, 5.095]	0 [(0
m0_pilot_s0	noise_0.5	500	15.216 [14.603, 15.801]	32.422 [30.932, 33.967]	12.905 [12.317, 13.458]	0 [(0
m0_pilot_s1	noise_0.5	500	13.618 [13.051, 14.229]	29.756 [28.343, 31.287]	11.825 [11.295, 12.395]	0 [(0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE8s	c 1 h
m0_pilot_s2	noise_0.5	500	14.817 [14.195, 15.438]	32.472 [31.078, 33.959]	12.928 [12.322, 13.541]	0 [(0
m1_pilot_s0	noise_0.5	500	6.117 [5.815, 6.419]	12.745 [11.928, 13.498]	5.074 [4.810, 5.356]	0 [(0
m1_pilot_s1	noise_0.5	500	5.961 [5.672, 6.250]	12.593 [11.846, 13.362]	4.948 [4.680, 5.218]	0 [(0
m1_pilot_s2	noise_0.5	500	5.697 [5.422, 5.965]	11.973 [11.245, 12.730]	4.692 [4.436, 4.943]	0 [(0
mind_pilot_s0	noise_0.5	500	15.551 [14.996, 16.110]	33.091 [31.724, 34.483]	12.881 [12.380, 13.374]	0 [(0
mind_pilot_s1	noise_0.5	500	15.329 [14.671, 15.989]	33.480 [31.925, 35.055]	13.213 [12.613, 13.817]	0 [(0
mind_pilot_s2	noise_0.5	500	14.537 [13.963, 15.165]	31.200 [29.783, 32.688]	12.949 [12.408, 13.520]	0 [(0
oracle_log	noise_0.5	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [1 n
rl_pilot_s0	noise_0.5	500	8.649 [8.262, 9.050]	18.649 [17.711, 19.600]	7.766 [7.387, 8.151]	0 [(0
rl_pilot_s1	noise_0.5	500	8.257 [7.874, 8.621]	18.006 [17.072, 18.982]	7.321 [6.955, 7.681]	0 [(0
rl_pilot_s2	noise_0.5	500	7.425 [7.081, 7.760]	16.275 [15.393, 17.162]	6.506 [6.165, 6.824]	0 [(0
select_m1_s0	noise_0.5	500	5.605 [5.320, 5.914]	12.045 [11.244, 12.806]	4.655 [4.388, 4.943]	0 [(0
select_m1_s1	noise_0.5	500	5.417 [5.132, 5.694]	11.605 [10.887, 12.307]	4.495 [4.235, 4.748]	0 [(0
select_m1_s2	noise_0.5	500	5.294 [5.012, 5.563]	11.303 [10.579, 12.026]	4.385 [4.134, 4.634]	0 [(0
cp	noise_1	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [(0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE8s	c 1 h
cv	noise_1	500	16.053 [15.473, 16.659]	35.912 [34.431, 37.395]	16.043 [15.459, 16.656]	0 [(0
extra_sft_pilot_s0	noise_1	500	9.516 [9.147, 9.893]	19.294 [18.378, 20.178]	8.081 [7.714, 8.431]	0 [(0
extra_sft_pilot_s1	noise_1	500	9.350 [8.998, 9.726]	18.761 [17.878, 19.662]	7.790 [7.451, 8.135]	0 [(0
extra_sft_pilot_s2	noise_1	500	9.039 [8.698, 9.408]	18.305 [17.455, 19.127]	7.665 [7.352, 8.000]	0 [(0
filtered_sft_pilot_s0	noise_1	500	9.563 [9.192, 9.945]	19.408 [18.513, 20.271]	8.100 [7.763, 8.450]	0 [(0
filtered_sft_pilot_s1	noise_1	500	9.434 [9.045, 9.844]	18.728 [17.851, 19.631]	7.990 [7.627, 8.365]	0 [(0
filtered_sft_pilot_s2	noise_1	500	9.088 [8.723, 9.461]	18.417 [17.567, 19.266]	7.786 [7.452, 8.135]	0 [(0
m0_pilot_s0	noise_1	500	17.987 [17.401, 18.594]	39.057 [37.527, 40.571]	15.042 [14.503, 15.614]	0 [(0
m0_pilot_s1	noise_1	500	16.976 [16.287, 17.658]	37.214 [35.472, 38.902]	15.249 [14.591, 15.887]	0 [(0
m0_pilot_s2	noise_1	500	20.223 [19.501, 20.935]	44.563 [42.744, 46.383]	17.208 [16.545, 17.857]	0 [(0
m1_pilot_s0	noise_1	500	9.510 [9.157, 9.871]	19.267 [18.371, 20.093]	8.003 [7.657, 8.349]	0 [(0
m1_pilot_s1	noise_1	500	9.284 [8.933, 9.662]	18.639 [17.698, 19.521]	7.641 [7.317, 7.984]	0 [(0
m1_pilot_s2	noise_1	500	9.252 [8.908, 9.645]	18.723 [17.863, 19.631]	7.785 [7.446, 8.140]	0 [(0
mind_pilot_s0	noise_1	500	19.390 [18.775, 20.027]	41.488 [39.837, 43.151]	15.785 [15.203, 16.406]	0 [(0
mind_pilot_s1	noise_1	500	18.175 [17.523, 18.869]	39.656 [37.971, 41.336]	15.686 [15.066, 16.328]	0 [(0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	h
mind_pilot_s2	noise_1	500	18.827 [18.161, 19.481]	40.571 [38.942, 42.322]	16.423 [15.772, 17.036]	0 [0
oracle_log	noise_1	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	noise_1	500	11.837 [11.368, 12.310]	24.176 [23.124, 25.247]	10.578 [10.129, 11.021]	0 [0
rl_pilot_s1	noise_1	500	11.122 [10.693, 11.576]	22.769 [21.726, 23.834]	9.781 [9.367, 10.212]	0 [0
rl_pilot_s2	noise_1	500	10.860 [10.422, 11.294]	22.758 [21.757, 23.804]	9.632 [9.195, 10.046]	0 [0
select_m1_s0	noise_1	500	8.572 [8.220, 8.941]	17.445 [16.558, 18.315]	7.212 [6.887, 7.544]	0 [0
select_m1_s1	noise_1	500	8.447 [8.077, 8.835]	17.133 [16.252, 18.038]	7.054 [6.695, 7.427]	0 [0
select_m1_s2	noise_1	500	8.332 [7.983, 8.716]	16.778 [15.949, 17.636]	7.109 [6.785, 7.454]	0 [0

Paired differences vs `m0_pilot_s0` (method − reference; group-level bootstrap 95% CI; groups = heuristic source-segment components, D-025):

method	condition	Δ joint minADE@68s	rel. Δ	Δ overlap rate (pp)	n
cp	clean	+22.966 [+21.252, +24.762]	+608.2% [+562.9%, +655.8%]	-14.56 [-16.43, -12.74]	500
cp	compound_0	+16.690 [+14.942, +18.595]	+166.0% [+148.6%, +185.0%]	-16.37 [-18.69, -14.03]	500
cp	missing_0.1	+22.801 [+21.119, +24.616]	+578.6% [+535.9%, +624.7%]	-15.03 [-17.00, -13.16]	500
cp	missing_0.3	+22.433 [+20.755, +24.223]	+520.6% [+481.6%, +562.1%]	-18.14 [-20.47, -16.13]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
cp	missing_0.5	+21.966 [+20.294, +23.759]	+459.9% [+424.9%, +497.5%]	-17.57 [-19.85, -15.46]	500
cp	missing_0.7	+20.922 [+19.265, +22.724]	+359.5% [+331.0%, +390.5%]	-19.77 [-22.20, -17.61]	500
cp	noise_0.1	+20.515 [+18.807, +22.335]	+329.5% [+302.0%, +358.7%]	-16.77 [-18.90, -14.50]	500
cp	noise_0.3	+14.324 [+12.581, +16.308]	+115.4% [+101.3%, +131.3%]	-13.13 [-14.99, -11.36]	500
cp	noise_0.5	+11.526 [+9.697, +13.513]	+75.8% [+63.7%, +88.8%]	-12.32 [-14.12, -10.63]	500
cp	noise_1	+8.755 [+6.867, +10.780]	+48.7% [+38.2%, +59.9%]	-11.96 [-13.84, -10.27]	500
cv	clean	+7.046 [+6.443, +7.673]	+186.6% [+170.6%, +203.2%]	+2.07 [-1.30, +5.41]	500
cv	compound_0	+1.815 [+1.092, +2.558]	+18.1% [+10.9%, +25.4%]	+1.47 [-1.70, +4.75]	500
cv	missing_0.1	+6.873 [+6.285, +7.475]	+174.4% [+159.5%, +189.7%]	+1.60 [-1.77, +5.06]	500
cv	missing_0.3	+6.459 [+5.858, +7.072]	+149.9% [+135.9%, +164.1%]	-1.90 [-5.25, +1.50]	500
cv	missing_0.5	+5.986 [+5.375, +6.584]	+125.3% [+112.5%, +137.9%]	-0.73 [-3.98, +2.43]	500
cv	missing_0.7	+5.607 [+4.986, +6.213]	+96.4% [+85.7%, +106.8%]	-3.54 [-6.83, -0.30]	500
cv	noise_0.1	+4.714 [+4.064, +5.388]	+75.7% [+65.3%, +86.5%]	+0.07 [-3.39, +3.65]	500
cv	noise_0.3	-0.786 [-1.519, +0.053]	-6.3% [-12.2%, +0.4%]	+4.51 [+1.03, +7.91]	500
cv	noise_0.5	-2.629 [-3.480, -1.774]	-17.3% [-22.9%, -11.7%]	+3.51 [+0.53, +6.81]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
cv	noise_1	-1.934 [-2.724, -1.139]	-10.8% [-15.1%, -6.3%]	+0.47 [-2.32, +3.03]	500
extra_sft_pilot_s0	clean	-0.322 [-0.506, -0.125]	-8.5% [-13.4%, -3.3%]	+1.27 [-0.74, +3.43]	500
extra_sft_pilot_s0	compound_0	-4.705 [-5.176, -4.242]	-46.8% [-51.5%, -42.2%]	+0.60 [-1.91, +3.23]	500
extra_sft_pilot_s0	missing_0.1	-0.501 [-0.700, -0.307]	-12.7% [-17.8%, -7.8%]	-0.53 [-2.47, +1.47]	500
extra_sft_pilot_s0	missing_0.3	-0.765 [-0.988, -0.558]	-17.8% [-22.9%, -12.9%]	-1.17 [-3.41, +0.93]	500
extra_sft_pilot_s0	missing_0.5	-1.139 [-1.408, -0.876]	-23.8% [-29.5%, -18.3%]	-1.27 [-3.51, +0.90]	500
extra_sft_pilot_s0	missing_0.7	-1.576 [-1.910, -1.288]	-27.1% [-32.8%, -22.1%]	-0.23 [-2.74, +2.27]	500
extra_sft_pilot_s0	noise_0.1	-2.108 [-2.438, -1.785]	-33.9% [-39.2%, -28.7%]	+1.50 [-0.97, +4.08]	500
extra_sft_pilot_s0	noise_0.3	-7.469 [-8.019, -6.897]	-60.1% [-64.6%, -55.5%]	+4.71 [+2.34, +7.03]	500
extra_sft_pilot_s0	noise_0.5	-9.092 [-9.710, -8.439]	-59.8% [-63.8%, -55.5%]	+3.61 [+1.30, +6.08]	500
extra_sft_pilot_s0	noise_1	-8.471 [-9.101, -7.884]	-47.1% [-50.6%, -43.8%]	+2.57 [+0.60, +4.52]	500
extra_sft_pilot_s1	clean	-0.306 [-0.519, -0.102]	-8.1% [-13.8%, -2.7%]	-1.00 [-3.04, +0.94]	500
extra_sft_pilot_s1	compound_0	-4.860 [-5.360, -4.375]	-48.3% [-53.3%, -43.5%]	+0.43 [-2.23, +3.17]	500
extra_sft_pilot_s1	missing_0.1	-0.429 [-0.645, -0.216]	-10.9% [-16.4%, -5.5%]	-0.67 [-2.83, +1.53]	500
extra_sft_pilot_s1	missing_0.3	-0.797 [-1.046, -0.567]	-18.5% [-24.3%, -13.1%]	-3.81 [-6.31, -1.37]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
extra_sft_pilot_s1	missing_0.5	-1.164 [-1.439, -0.906]	-24.4% [-30.1%, -19.0%]	-2.30 [-4.80, +0.10]	500
extra_sft_pilot_s1	missing_0.7	-1.669 [-2.004, -1.349]	-28.7% [-34.4%, -23.2%]	-3.01 [-5.63, -0.34]	500
extra_sft_pilot_s1	noise_0.1	-2.220 [-2.580, -1.894]	-35.7% [-41.4%, -30.4%]	+0.03 [-2.38, +2.58]	500
extra_sft_pilot_s1	noise_0.3	-7.666 [-8.219, -7.088]	-61.7% [-66.2%, -57.1%]	+4.28 [+1.57, +6.78]	500
extra_sft_pilot_s1	noise_0.5	-9.369 [-10.000, -8.729]	-61.6% [-65.7%, -57.4%]	+3.47 [+1.10, +5.75]	500
extra_sft_pilot_s1	noise_1	-8.637 [-9.301, -8.033]	-48.0% [-51.7%, -44.7%]	+3.41 [+1.41, +5.40]	500
extra_sft_pilot_s2	clean	-0.283 [-0.462, -0.105]	-7.5% [-12.2%, -2.8%]	-0.20 [-2.23, +1.81]	500
extra_sft_pilot_s2	compound_0	-4.731 [-5.208, -4.277]	-47.1% [-51.8%, -42.5%]	+0.84 [-1.74, +3.55]	500
extra_sft_pilot_s2	missing_0.1	-0.328 [-0.523, -0.131]	-8.3% [-13.3%, -3.3%]	-0.43 [-2.66, +1.67]	500
extra_sft_pilot_s2	missing_0.3	-0.834 [-1.072, -0.610]	-19.4% [-24.9%, -14.2%]	-3.51 [-5.88, -1.20]	500
extra_sft_pilot_s2	missing_0.5	-0.984 [-1.252, -0.727]	-20.6% [-26.2%, -15.2%]	-4.24 [-6.45, -1.97]	500
extra_sft_pilot_s2	missing_0.7	-1.657 [-2.001, -1.320]	-28.5% [-34.4%, -22.7%]	-3.04 [-5.72, -0.60]	500
extra_sft_pilot_s2	noise_0.1	-2.104 [-2.441, -1.772]	-33.8% [-39.2%, -28.5%]	-0.30 [-2.80, +2.40]	500
extra_sft_pilot_s2	noise_0.3	-7.555 [-8.124, -6.988]	-60.8% [-65.4%, -56.3%]	+5.91 [+3.27, +8.48]	500
extra_sft_pilot_s2	noise_0.5	-9.580 [-10.221, -8.944]	-63.0% [-67.2%, -58.8%]	+3.97 [+1.54, +6.51]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
extra_sft_pilot_s2	noise_1	-8.948 [-9.566, -8.348]	-49.7% [-53.2%, -46.4%]	+2.94 [+0.96, +4.92]	500
filtered_sft_pilot_s0	clean	-0.167 [-0.353, +0.017]	-4.4% [-9.4%, +0.4%]	+0.67 [-1.14, +2.74]	500
filtered_sft_pilot_s0	compound_0	-4.637 [-5.106, -4.185]	-46.1% [-50.8%, -41.6%]	+0.23 [-2.40, +2.98]	500
filtered_sft_pilot_s0	missing_0.1	-0.417 [-0.607, -0.227]	-10.6% [-15.4%, -5.8%]	+0.33 [-1.47, +2.27]	500
filtered_sft_pilot_s0	missing_0.3	-0.782 [-1.004, -0.565]	-18.1% [-23.3%, -13.1%]	-2.44 [-4.64, -0.36]	500
filtered_sft_pilot_s0	missing_0.5	-1.022 [-1.285, -0.768]	-21.4% [-26.9%, -16.1%]	-2.74 [-5.01, -0.57]	500
filtered_sft_pilot_s0	missing_0.7	-1.319 [-1.636, -1.010]	-22.7% [-28.1%, -17.4%]	-3.01 [-5.46, -0.64]	500
filtered_sft_pilot_s0	noise_0.1	-2.142 [-2.480, -1.811]	-34.4% [-39.8%, -29.1%]	+0.47 [-2.04, +3.23]	500
filtered_sft_pilot_s0	noise_0.3	-7.313 [-7.837, -6.751]	-58.9% [-63.1%, -54.4%]	+3.34 [+0.83, +5.71]	500
filtered_sft_pilot_s0	noise_0.5	-8.938 [-9.550, -8.286]	-58.7% [-62.8%, -54.5%]	+3.24 [+0.90, +5.58]	500
filtered_sft_pilot_s0	noise_1	-8.424 [-9.018, -7.848]	-46.8% [-50.1%, -43.6%]	+2.47 [+0.44, +4.48]	500
filtered_sft_pilot_s1	clean	-0.217 [-0.418, -0.011]	-5.7% [-11.1%, -0.3%]	-0.27 [-2.31, +1.77]	500
filtered_sft_pilot_s1	compound_0	-4.429 [-4.922, -3.961]	-44.1% [-49.0%, -39.4%]	-2.07 [-4.73, +0.50]	500
filtered_sft_pilot_s1	missing_0.1	-0.351 [-0.554, -0.142]	-8.9% [-14.1%, -3.6%]	-2.17 [-4.29, -0.10]	500
filtered_sft_pilot_s1	missing_0.3	-0.725 [-0.959, -0.495]	-16.8% [-22.3%, -11.5%]	-4.07 [-6.55, -1.74]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
filtered_sft_pilot_s1	missing_0.5	-1.080 [-1.358, -0.824]	-22.6% [-28.4%, -17.3%]	-3.47 [-5.94, -1.04]	500
filtered_sft_pilot_s1	missing_0.7	-1.375 [-1.705, -1.047]	-23.6% [-29.3%, -18.0%]	-3.41 [-6.16, -0.80]	500
filtered_sft_pilot_s1	noise_0.1	-2.112 [-2.450, -1.783]	-33.9% [-39.3%, -28.6%]	-1.44 [-3.97, +1.17]	500
filtered_sft_pilot_s1	noise_0.3	-7.318 [-7.859, -6.759]	-58.9% [-63.3%, -54.4%]	+3.64 [+1.16, +6.15]	500
filtered_sft_pilot_s1	noise_0.5	-8.965 [-9.589, -8.328]	-58.9% [-63.0%, -54.7%]	+1.87 [-0.53, +4.12]	500
filtered_sft_pilot_s1	noise_1	-8.553 [-9.169, -7.944]	-47.6% [-51.0%, -44.2%]	+4.44 [+2.16, +6.64]	500
filtered_sft_pilot_s2	clean	-0.288 [-0.483, -0.097]	-7.6% [-12.8%, -2.6%]	-0.84 [-2.99, +1.23]	500
filtered_sft_pilot_s2	compound_0	-4.613 [-5.097, -4.154]	-45.9% [-50.7%, -41.3%]	+0.57 [-2.04, +3.37]	500
filtered_sft_pilot_s2	missing_0.1	-0.396 [-0.597, -0.192]	-10.1% [-15.2%, -4.9%]	-0.63 [-2.83, +1.44]	500
filtered_sft_pilot_s2	missing_0.3	-0.714 [-0.936, -0.503]	-16.6% [-21.7%, -11.7%]	-3.57 [-6.05, -1.14]	500
filtered_sft_pilot_s2	missing_0.5	-0.992 [-1.249, -0.731]	-20.8% [-26.2%, -15.3%]	-4.21 [-6.67, -1.81]	500
filtered_sft_pilot_s2	missing_0.7	-1.406 [-1.740, -1.076]	-24.2% [-29.9%, -18.5%]	-3.74 [-6.15, -1.31]	500
filtered_sft_pilot_s2	noise_0.1	-2.109 [-2.453, -1.766]	-33.9% [-39.4%, -28.4%]	-1.17 [-3.59, +1.36]	500
filtered_sft_pilot_s2	noise_0.3	-7.411 [-7.959, -6.831]	-59.7% [-64.1%, -55.0%]	+3.41 [+0.80, +5.78]	500
filtered_sft_pilot_s2	noise_0.5	-9.446 [-10.040, -8.826]	-62.1% [-66.0%, -58.0%]	+1.84 [-0.37, +4.08]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
filtered_sft_pilot_s2	noise_1	-8.899 [-9.489, -8.333]	-49.5% [-52.8%, -46.3%]	+2.10 [+0.07, +4.07]	500
m0_pilot_s1	clean	-0.110 [-0.323, +0.103]	-2.9% [-8.5%, +2.7%]	+0.63 [-1.31, +2.61]	500
m0_pilot_s1	compound_0	+0.352 [-0.168, +0.900]	+3.5% [-1.7%, +9.0%]	-0.53 [-2.84, +1.79]	500
m0_pilot_s1	missing_0.1	-0.156 [-0.356, +0.055]	-4.0% [-9.0%, +1.4%]	+1.00 [-1.10, +3.05]	500
m0_pilot_s1	missing_0.3	+0.039 [-0.221, +0.297]	+0.9% [-5.1%, +6.9%]	-0.50 [-2.87, +1.81]	500
m0_pilot_s1	missing_0.5	+0.161 [-0.142, +0.453]	+3.4% [-3.0%, +9.5%]	+2.77 [+0.40, +5.21]	500
m0_pilot_s1	missing_0.7	+0.392 [+0.029, +0.763]	+6.7% [+0.5%, +13.1%]	+1.10 [-1.47, +3.68]	500
m0_pilot_s1	noise_0.1	-0.161 [-0.481, +0.186]	-2.6% [-7.7%, +3.0%]	-0.10 [-2.54, +2.38]	500
m0_pilot_s1	noise_0.3	-1.323 [-1.850, -0.776]	-10.7% [-14.9%, -6.2%]	+0.70 [-1.66, +2.92]	500
m0_pilot_s1	noise_0.5	-1.598 [-2.210, -0.952]	-10.5% [-14.5%, -6.3%]	-2.04 [-4.07, +0.07]	500
m0_pilot_s1	noise_1	-1.011 [-1.759, -0.286]	-5.6% [-9.8%, -1.6%]	+0.10 [-1.67, +1.94]	500
m0_pilot_s2	clean	-0.010 [-0.217, +0.187]	-0.3% [-5.8%, +4.9%]	+0.80 [-1.20, +2.83]	500
m0_pilot_s2	compound_0	+0.220 [-0.276, +0.704]	+2.2% [-2.7%, +7.0%]	-1.64 [-3.88, +0.74]	500
m0_pilot_s2	missing_0.1	+0.009 [-0.176, +0.191]	+0.2% [-4.5%, +4.9%]	+2.47 [+0.33, +4.45]	500
m0_pilot_s2	missing_0.3	-0.200 [-0.448, +0.040]	-4.6% [-10.4%, +0.9%]	+0.70 [-1.78, +3.05]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
m0_pilot_s2	missing_0.5	-0.146 [-0.435, +0.132]	-3.1% [-9.1%, +2.8%]	+4.18 [+1.74, +6.75]	500
m0_pilot_s2	missing_0.7	+0.015 [-0.315, +0.357]	+0.3% [-5.4%, +6.1%]	+4.01 [+1.17, +6.59]	500
m0_pilot_s2	noise_0.1	-0.516 [-0.869, -0.155]	-8.3% [-14.0%, -2.5%]	-1.20 [-3.81, +1.33]	500
m0_pilot_s2	noise_0.3	-0.989 [-1.556, -0.401]	-8.0% [-12.5%, -3.2%]	+1.37 [-1.07, +3.70]	500
m0_pilot_s2	noise_0.5	-0.398 [-1.080, +0.345]	-2.6% [-7.1%, +2.3%]	+0.27 [-1.94, +2.50]	500
m0_pilot_s2	noise_1	+2.236 [+1.458, +3.017]	+12.4% [+8.1%, +16.8%]	+1.00 [-0.80, +2.77]	500
m1_pilot_s0	clean	-0.214 [-0.391, -0.036]	-5.7% [-10.3%, -0.9%]	+1.27 [-0.53, +3.21]	500
m1_pilot_s0	compound_0	-4.718 [-5.187, -4.238]	-46.9% [-51.6%, -42.2%]	+0.40 [-2.11, +3.02]	500
m1_pilot_s0	missing_0.1	-0.479 [-0.657, -0.295]	-12.2% [-16.7%, -7.5%]	+0.10 [-1.87, +2.10]	500
m1_pilot_s0	missing_0.3	-0.716 [-0.934, -0.512]	-16.6% [-21.7%, -11.9%]	-2.47 [-4.71, -0.30]	500
m1_pilot_s0	missing_0.5	-1.118 [-1.377, -0.862]	-23.4% [-28.8%, -18.1%]	-2.87 [-5.22, -0.57]	500
m1_pilot_s0	missing_0.7	-1.612 [-1.931, -1.318]	-27.7% [-33.2%, -22.7%]	-1.17 [-3.64, +1.24]	500
m1_pilot_s0	noise_0.1	-2.021 [-2.365, -1.688]	-32.5% [-38.0%, -27.1%]	+1.30 [-1.11, +3.83]	500
m1_pilot_s0	noise_0.3	-7.478 [-8.014, -6.927]	-60.2% [-64.5%, -55.8%]	+4.18 [+1.64, +6.59]	500
m1_pilot_s0	noise_0.5	-9.099 [-9.698, -8.479]	-59.8% [-63.7%, -55.7%]	+3.57 [+1.27, +5.99]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
m1_pilot_s0	noise_1	-8.477 [-9.066, -7.874]	-47.1% [-50.4%, -43.8%]	+3.37 [+1.21, +5.43]	500
m1_pilot_s1	clean	-0.295 [-0.498, -0.106]	-7.8% [-13.2%, -2.8%]	-0.33 [-2.35, +1.71]	500
m1_pilot_s1	compound_0	-4.788 [-5.277, -4.299]	-47.6% [-52.5%, -42.8%]	+0.23 [-2.33, +3.03]	500
m1_pilot_s1	missing_0.1	-0.408 [-0.612, -0.194]	-10.3% [-15.5%, -4.9%]	-0.30 [-2.46, +1.81]	500
m1_pilot_s1	missing_0.3	-0.807 [-1.037, -0.574]	-18.7% [-24.1%, -13.3%]	-1.10 [-3.61, +1.27]	500
m1_pilot_s1	missing_0.5	-1.164 [-1.437, -0.897]	-24.4% [-30.1%, -18.8%]	-0.94 [-3.40, +1.54]	500
m1_pilot_s1	missing_0.7	-1.601 [-1.950, -1.275]	-27.5% [-33.5%, -21.9%]	-1.14 [-3.67, +1.34]	500
m1_pilot_s1	noise_0.1	-2.283 [-2.635, -1.959]	-36.7% [-42.3%, -31.5%]	+1.27 [-1.37, +3.99]	500
m1_pilot_s1	noise_0.3	-7.561 [-8.113, -6.986]	-60.9% [-65.3%, -56.3%]	+4.74 [+2.06, +7.34]	500
m1_pilot_s1	noise_0.5	-9.254 [-9.880, -8.623]	-60.8% [-64.9%, -56.7%]	+3.67 [+1.34, +6.10]	500
m1_pilot_s1	noise_1	-8.703 [-9.324, -8.099]	-48.4% [-51.8%, -45.0%]	+1.80 [-0.03, +3.79]	500
m1_pilot_s2	clean	-0.281 [-0.472, -0.103]	-7.4% [-12.5%, -2.7%]	+0.17 [-2.07, +2.41]	500
m1_pilot_s2	compound_0	-4.788 [-5.262, -4.331]	-47.6% [-52.4%, -43.1%]	+0.37 [-2.27, +3.11]	500
m1_pilot_s2	missing_0.1	-0.477 [-0.660, -0.298]	-12.1% [-16.8%, -7.6%]	-0.20 [-2.34, +1.87]	500
m1_pilot_s2	missing_0.3	-0.786 [-0.990, -0.587]	-18.2% [-23.0%, -13.6%]	-2.47 [-4.92, -0.17]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
m1_pilot_s2	missing_0.5	-1.013 [-1.279, -0.766]	-21.2% [-26.8%, -16.0%]	-2.34 [-4.77, +0.13]	500
m1_pilot_s2	missing_0.7	-1.625 [-1.954, -1.292]	-27.9% [-33.6%, -22.2%]	-2.67 [-5.16, -0.20]	500
m1_pilot_s2	noise_0.1	-2.194 [-2.527, -1.863]	-35.2% [-40.6%, -29.9%]	+0.20 [-2.24, +2.93]	500
m1_pilot_s2	noise_0.3	-7.586 [-8.129, -7.007]	-61.1% [-65.5%, -56.4%]	+4.18 [+1.64, +6.51]	500
m1_pilot_s2	noise_0.5	-9.518 [-10.155, -8.876]	-62.6% [-66.7%, -58.3%]	+3.57 [+1.17, +5.91]	500
m1_pilot_s2	noise_1	-8.735 [-9.337, -8.131]	-48.6% [-51.9%, -45.2%]	+2.51 [+0.60, +4.38]	500
mind_pilot_s0	clean	+0.095 [-0.104, +0.289]	+2.5% [-2.7%, +7.7%]	+3.14 [+1.07, +5.32]	500
mind_pilot_s0	compound_0	+0.303 [-0.150, +0.736]	+3.0% [-1.5%, +7.3%]	-1.14 [-3.16, +0.90]	500
mind_pilot_s0	missing_0.1	+0.098 [-0.100, +0.294]	+2.5% [-2.5%, +7.5%]	+3.84 [+1.54, +5.98]	500
mind_pilot_s0	missing_0.3	+0.124 [-0.121, +0.368]	+2.9% [-2.8%, +8.5%]	+1.47 [-1.00, +3.77]	500
mind_pilot_s0	missing_0.5	+0.372 [+0.093, +0.653]	+7.8% [+1.9%, +13.7%]	+3.24 [+1.00, +5.62]	500
mind_pilot_s0	missing_0.7	+0.320 [-0.009, +0.624]	+5.5% [-0.1%, +10.7%]	+1.04 [-1.36, +3.55]	500
mind_pilot_s0	noise_0.1	+0.374 [+0.036, +0.696]	+6.0% [+0.6%, +11.2%]	+1.34 [-0.90, +3.64]	500
mind_pilot_s0	noise_0.3	+0.382 [-0.094, +0.901]	+3.1% [-0.8%, +7.3%]	+0.20 [-1.70, +2.18]	500
mind_pilot_s0	noise_0.5	+0.336 [-0.291, +0.944]	+2.2% [-1.9%, +6.2%]	-0.50 [-2.31, +1.37]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
mind_pilot_s0	noise_1	+1.403 [+0.749, +2.120]	+7.8% [+4.2%, +11.8%]	+0.23 [-1.60, +1.98]	500
mind_pilot_s1	clean	+0.245 [+0.022, +0.463]	+6.5% [+0.6%, +12.3%]	+3.81 [+1.63, +5.98]	500
mind_pilot_s1	compound_0	+1.271 [+0.739, +1.799]	+12.6% [+7.4%, +17.9%]	-2.34 [-4.61, -0.10]	500
mind_pilot_s1	missing_0.1	+0.135 [-0.073, +0.352]	+3.4% [-1.8%, +8.9%]	+2.84 [+0.53, +5.02]	500
mind_pilot_s1	missing_0.3	+0.479 [+0.220, +0.740]	+11.1% [+5.1%, +17.2%]	-0.10 [-2.61, +2.38]	500
mind_pilot_s1	missing_0.5	+0.688 [+0.388, +1.013]	+14.4% [+8.1%, +21.2%]	+2.67 [+0.23, +5.20]	500
mind_pilot_s1	missing_0.7	+1.250 [+0.876, +1.643]	+21.5% [+15.1%, +28.2%]	-1.00 [-3.61, +1.34]	500
mind_pilot_s1	noise_0.1	+0.001 [-0.359, +0.369]	+0.0% [-5.8%, +5.9%]	+0.80 [-1.74, +3.28]	500
mind_pilot_s1	noise_0.3	+0.048 [-0.484, +0.629]	+0.4% [-3.9%, +5.1%]	-1.70 [-3.94, +0.44]	500
mind_pilot_s1	noise_0.5	+0.114 [-0.571, +0.832]	+0.7% [-3.8%, +5.5%]	-2.04 [-3.87, -0.13]	500
mind_pilot_s1	noise_1	+0.188 [-0.483, +0.907]	+1.0% [-2.7%, +5.0%]	-1.30 [-3.03, +0.47]	500
mind_pilot_s2	clean	+0.021 [-0.188, +0.222]	+0.5% [-5.0%, +5.9%]	+5.71 [+3.48, +7.93]	500
mind_pilot_s2	compound_0	-0.243 [-0.757, +0.269]	-2.4% [-7.5%, +2.7%]	+0.90 [-1.84, +3.70]	500
mind_pilot_s2	missing_0.1	+0.232 [+0.013, +0.454]	+5.9% [+0.3%, +11.5%]	+5.91 [+3.76, +8.25]	500
mind_pilot_s2	missing_0.3	+0.223 [-0.008, +0.461]	+5.2% [-0.2%, +10.7%]	+5.44 [+2.95, +8.03]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
mind_pilot_s2	missing_0.5	+0.145 [-0.156, +0.439]	+3.0% [-3.3%, +9.2%]	+6.11 [+3.59, +8.55]	500
mind_pilot_s2	missing_0.7	+0.490 [+0.140, +0.842]	+8.4% [+2.4%, +14.5%]	+4.11 [+1.44, +6.73]	500
mind_pilot_s2	noise_0.1	-0.691 [-1.075, -0.323]	-11.1% [-17.3%, -5.2%]	+2.54 [-0.00, +5.19]	500
mind_pilot_s2	noise_0.3	-1.384 [-1.955, -0.766]	-11.1% [-15.7%, -6.2%]	+2.34 [+0.00, +4.80]	500
mind_pilot_s2	noise_0.5	-0.678 [-1.371, +0.032]	-4.5% [-9.0%, +0.2%]	-0.33 [-2.47, +1.81]	500
mind_pilot_s2	noise_1	+0.840 [+0.083, +1.572]	+4.7% [+0.5%, +8.7%]	+0.73 [-1.26, +2.78]	500
oracle_log	clean	-3.776 [-4.037, -3.522]	-100.0% [-106.9%, -93.3%]	+nan [+nan, +nan]	500
oracle_log	compound_0	-10.052 [-10.533, -9.579]	-100.0% [-104.8%, -95.3%]	+nan [+nan, +nan]	500
oracle_log	missing_0.1	-3.941 [-4.195, -3.691]	-100.0% [-106.4%, -93.7%]	+nan [+nan, +nan]	500
oracle_log	missing_0.3	-4.309 [-4.591, -4.041]	-100.0% [-106.5%, -93.8%]	+nan [+nan, +nan]	500
oracle_log	missing_0.5	-4.776 [-5.097, -4.461]	-100.0% [-106.7%, -93.4%]	+nan [+nan, +nan]	500
oracle_log	missing_0.7	-5.819 [-6.233, -5.424]	-100.0% [-107.1%, -93.2%]	+nan [+nan, +nan]	500
oracle_log	noise_0.1	-6.226 [-6.606, -5.849]	-100.0% [-106.1%, -93.9%]	+nan [+nan, +nan]	500
oracle_log	noise_0.3	-12.417 [-12.942, -11.855]	-100.0% [-104.2%, -95.5%]	+nan [+nan, +nan]	500
oracle_log	noise_0.5	-15.216 [-15.801, -14.603]	-100.0% [-103.8%, -96.0%]	+nan [+nan, +nan]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
oracle_log	noise_1	-17.987 [-18.594, -17.401]	-100.0% [-103.4%, -96.7%]	+nan [+nan, +nan]	500
rl_pilot_s0	clean	+0.729 [+0.520, +0.944]	+19.3% [+13.8%, +25.0%]	-9.29 [-11.01, -7.48]	500
rl_pilot_s0	compound_0	-2.336 [-2.829, -1.875]	-23.2% [-28.1%, -18.7%]	-9.08 [-11.34, -6.69]	500
rl_pilot_s0	missing_0.1	+0.544 [+0.317, +0.780]	+13.8% [+8.0%, +19.8%]	-10.62 [-12.52, -8.75]	500
rl_pilot_s0	missing_0.3	+0.182 [-0.082, +0.445]	+4.2% [-1.9%, +10.3%]	-13.33 [-15.66, -11.20]	500
rl_pilot_s0	missing_0.5	+0.199 [-0.113, +0.499]	+4.2% [-2.4%, +10.4%]	-13.66 [-16.06, -11.38]	500
rl_pilot_s0	missing_0.7	-0.293 [-0.641, +0.067]	-5.0% [-11.0%, +1.2%]	-14.93 [-17.43, -12.61]	500
rl_pilot_s0	noise_0.1	-0.605 [-0.948, -0.251]	-9.7% [-15.2%, -4.0%]	-11.26 [-13.39, -9.00]	500
rl_pilot_s0	noise_0.3	-5.194 [-5.689, -4.660]	-41.8% [-45.8%, -37.5%]	-5.81 [-7.92, -3.71]	500
rl_pilot_s0	noise_0.5	-6.567 [-7.187, -5.924]	-43.2% [-47.2%, -38.9%]	-4.04 [-6.00, -2.10]	500
rl_pilot_s0	noise_1	-6.150 [-6.771, -5.540]	-34.2% [-37.6%, -30.8%]	-2.67 [-4.57, -0.80]	500
rl_pilot_s1	clean	+0.634 [+0.412, +0.853]	+16.8% [+10.9%, +22.6%]	-10.29 [-12.22, -8.35]	500
rl_pilot_s1	compound_0	-2.558 [-3.051, -2.068]	-25.4% [-30.4%, -20.6%]	-9.22 [-11.58, -6.99]	500
rl_pilot_s1	missing_0.1	+0.487 [+0.257, +0.728]	+12.4% [+6.5%, +18.5%]	-10.72 [-12.86, -8.66]	500
rl_pilot_s1	missing_0.3	+0.110 [-0.168, +0.384]	+2.6% [-3.9%, +8.9%]	-13.93 [-16.37, -11.78]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
rl_pilot_s1	missing_0.5	-0.010 [-0.307, +0.282]	-0.2% [-6.4%, +5.9%]	-13.19 [-15.57, -10.95]	500
rl_pilot_s1	missing_0.7	-0.167 [-0.536, +0.187]	-2.9% [-9.2%, +3.2%]	-13.66 [-16.13, -11.30]	500
rl_pilot_s1	noise_0.1	-0.624 [-0.986, -0.287]	-10.0% [-15.8%, -4.6%]	-11.76 [-14.09, -9.42]	500
rl_pilot_s1	noise_0.3	-5.413 [-5.958, -4.828]	-43.6% [-48.0%, -38.9%]	-6.11 [-8.42, -3.90]	500
rl_pilot_s1	noise_0.5	-6.958 [-7.602, -6.316]	-45.7% [-50.0%, -41.5%]	-4.44 [-6.37, -2.40]	500
rl_pilot_s1	noise_1	-6.865 [-7.483, -6.244]	-38.2% [-41.6%, -34.7%]	-2.44 [-4.25, -0.74]	500
rl_pilot_s2	clean	+0.364 [+0.139, +0.588]	+9.6% [+3.7%, +15.6%]	-8.32 [-10.22, -6.36]	500
rl_pilot_s2	compound_0	-3.286 [-3.778, -2.795]	-32.7% [-37.6%, -27.8%]	-8.92 [-11.29, -6.54]	500
rl_pilot_s2	missing_0.1	+0.256 [+0.045, +0.476]	+6.5% [+1.1%, +12.1%]	-9.35 [-11.38, -7.29]	500
rl_pilot_s2	missing_0.3	-0.086 [-0.340, +0.171]	-2.0% [-7.9%, +4.0%]	-12.16 [-14.50, -10.00]	500
rl_pilot_s2	missing_0.5	-0.397 [-0.669, -0.135]	-8.3% [-14.0%, -2.8%]	-11.76 [-14.06, -9.49]	500
rl_pilot_s2	missing_0.7	-0.769 [-1.128, -0.431]	-13.2% [-19.4%, -7.4%]	-13.09 [-15.57, -10.79]	500
rl_pilot_s2	noise_0.1	-1.177 [-1.512, -0.842]	-18.9% [-24.3%, -13.5%]	-10.62 [-13.01, -8.15]	500
rl_pilot_s2	noise_0.3	-6.119 [-6.678, -5.519]	-49.3% [-53.8%, -44.4%]	-4.78 [-7.06, -2.64]	500
rl_pilot_s2	noise_0.5	-7.790 [-8.401, -7.149]	-51.2% [-55.2%, -47.0%]	-3.71 [-5.87, -1.53]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
rl_pilot_s2	noise_1	-7.127 [-7.774, -6.480]	-39.6% [-43.2%, -36.0%]	-1.40 [-3.43, +0.64]	500
select_m1_s0	clean	-0.445 [-0.629, -0.259]	-11.8% [-16.7%, -6.9%]	-13.49 [-15.36, -11.72]	500
select_m1_s0	compound_0	-5.051 [-5.539, -4.584]	-50.2% [-55.1%, -45.6%]	-15.23 [-17.47, -13.03]	500
select_m1_s0	missing_0.1	-0.556 [-0.760, -0.341]	-14.1% [-19.3%, -8.7%]	-13.93 [-15.90, -12.11]	500
select_m1_s0	missing_0.3	-0.861 [-1.085, -0.648]	-20.0% [-25.2%, -15.0%]	-16.97 [-19.28, -14.96]	500
select_m1_s0	missing_0.5	-1.209 [-1.469, -0.947]	-25.3% [-30.8%, -19.8%]	-16.60 [-18.80, -14.60]	500
select_m1_s0	missing_0.7	-1.830 [-2.162, -1.522]	-31.5% [-37.1%, -26.2%]	-18.97 [-21.44, -16.70]	500
select_m1_s0	noise_0.1	-2.376 [-2.726, -2.041]	-38.2% [-43.8%, -32.8%]	-16.33 [-18.50, -14.11]	500
select_m1_s0	noise_0.3	-7.743 [-8.297, -7.198]	-62.4% [-66.8%, -58.0%]	-12.73 [-14.60, -10.98]	500
select_m1_s0	noise_0.5	-9.611 [-10.226, -8.963]	-63.2% [-67.2%, -58.9%]	-11.66 [-13.45, -9.86]	500
select_m1_s0	noise_1	-9.415 [-10.014, -8.808]	-52.3% [-55.7%, -49.0%]	-10.92 [-12.59, -9.28]	500
select_m1_s1	clean	-0.503 [-0.713, -0.304]	-13.3% [-18.9%, -8.0%]	-13.99 [-15.77, -12.16]	500
select_m1_s1	compound_0	-4.939 [-5.444, -4.459]	-49.1% [-54.2%, -44.4%]	-15.06 [-17.40, -12.84]	500
select_m1_s1	missing_0.1	-0.651 [-0.862, -0.435]	-16.5% [-21.9%, -11.0%]	-14.43 [-16.37, -12.58]	500
select_m1_s1	missing_0.3	-0.883 [-1.124, -0.652]	-20.5% [-26.1%, -15.1%]	-17.03 [-19.24, -15.07]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
select_m1_s1	missing_0.5	-1.265 [-1.529, -1.011]	-26.5% [-32.0%, -21.2%]	-16.30 [-18.44, -14.26]	500
select_m1_s1	missing_0.7	-1.687 [-2.018, -1.361]	-29.0% [-34.7%, -23.4%]	-18.20 [-20.55, -16.07]	500
select_m1_s1	noise_0.1	-2.335 [-2.687, -1.997]	-37.5% [-43.2%, -32.1%]	-16.30 [-18.47, -14.07]	500
select_m1_s1	noise_0.3	-7.806 [-8.355, -7.216]	-62.9% [-67.3%, -58.1%]	-12.59 [-14.59, -10.82]	500
select_m1_s1	noise_0.5	-9.798 [-10.397, -9.162]	-64.4% [-68.3%, -60.2%]	-11.92 [-13.72, -10.25]	500
select_m1_s1	noise_1	-9.540 [-10.173, -8.922]	-53.0% [-56.6%, -49.6%]	-11.22 [-12.93, -9.57]	500
select_m1_s2	clean	-0.442 [-0.632, -0.249]	-11.7% [-16.7%, -6.6%]	-12.73 [-14.67, -10.80]	500
select_m1_s2	compound_0	-4.946 [-5.453, -4.490]	-49.2% [-54.3%, -44.7%]	-14.76 [-17.07, -12.54]	500
select_m1_s2	missing_0.1	-0.544 [-0.732, -0.355]	-13.8% [-18.6%, -9.0%]	-13.69 [-15.74, -11.72]	500
select_m1_s2	missing_0.3	-0.758 [-0.997, -0.521]	-17.6% [-23.1%, -12.1%]	-17.10 [-19.44, -14.99]	500
select_m1_s2	missing_0.5	-1.206 [-1.460, -0.964]	-25.3% [-30.6%, -20.2%]	-16.40 [-18.64, -14.29]	500
select_m1_s2	missing_0.7	-1.878 [-2.202, -1.560]	-32.3% [-37.8%, -26.8%]	-18.50 [-20.85, -16.33]	500
select_m1_s2	noise_0.1	-2.142 [-2.500, -1.814]	-34.4% [-40.1%, -29.1%]	-15.97 [-18.10, -13.73]	500
select_m1_s2	noise_0.3	-7.860 [-8.411, -7.276]	-63.3% [-67.7%, -58.6%]	-12.73 [-14.63, -10.97]	500
select_m1_s2	noise_0.5	-9.922 [-10.555, -9.296]	-65.2% [-69.4%, -61.1%]	-11.69 [-13.44, -9.99]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
select_m1_s2	noise_1	-9.655 [-10.275, -9.033]	-53.7% [-57.1%, -50.2%]	-10.65 [-12.32, -9.02]	500
cp	CORE_SUITE	+18.928 [+17.208, +20.764]	+242.2% [+220.2%, +265.7%]	-15.49 [-16.74, -14.36]	500
cv	CORE_SUITE	+3.436 [+2.829, +4.064]	+44.0% [+36.2%, +52.0%]	+1.17 [-1.11, +3.57]	500
extra_sft_pilot_s0	CORE_SUITE	-3.512 [-3.720, -3.295]	-44.9% [-47.6%, -42.2%]	+1.14 [-0.07, +2.40]	500
extra_sft_pilot_s1	CORE_SUITE	-3.608 [-3.833, -3.382]	-46.2% [-49.0%, -43.3%]	+0.17 [-1.20, +1.53]	500
extra_sft_pilot_s2	CORE_SUITE	-3.564 [-3.780, -3.347]	-45.6% [-48.4%, -42.8%]	+0.23 [-1.12, +1.58]	500
filtered_sft_pilot_s0	CORE_SUITE	-3.435 [-3.632, -3.235]	-44.0% [-46.5%, -41.4%]	+0.37 [-0.90, +1.61]	500
filtered_sft_pilot_s1	CORE_SUITE	-3.425 [-3.639, -3.207]	-43.8% [-46.6%, -41.0%]	-0.94 [-2.34, +0.37]	500
filtered_sft_pilot_s2	CORE_SUITE	-3.511 [-3.716, -3.295]	-44.9% [-47.5%, -42.2%]	-0.72 [-2.11, +0.69]	500
m0_pilot_s1	CORE_SUITE	-0.506 [-0.704, -0.293]	-6.5% [-9.0%, -3.8%]	+0.31 [-0.81, +1.38]	500
m0_pilot_s2	CORE_SUITE	-0.373 [-0.581, -0.163]	-4.8% [-7.4%, -2.1%]	+1.30 [+0.17, +2.45]	500
m1_pilot_s0	CORE_SUITE	-3.485 [-3.698, -3.270]	-44.6% [-47.3%, -41.9%]	+0.63 [-0.66, +1.94]	500
m1_pilot_s1	CORE_SUITE	-3.580 [-3.799, -3.358]	-45.8% [-48.6%, -43.0%]	+1.22 [-0.19, +2.62]	500
m1_pilot_s2	CORE_SUITE	-3.596 [-3.810, -3.381]	-46.0% [-48.8%, -43.3%]	+0.49 [-0.85, +1.86]	500
mind_pilot_s0	CORE_SUITE	+0.281 [+0.107, +0.465]	+3.6% [+1.4%, +5.9%]	+1.60 [+0.55, +2.63]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
mind_pilot_s1	CORE_SUITE	+0.244 [+0.042, +0.457]	+3.1% [+0.5%, +5.8%]	+0.41 [-0.70, +1.46]	500
mind_pilot_s2	CORE_SUITE	-0.359 [-0.570, -0.136]	-4.6% [-7.3%, -1.7%]	+3.67 [+2.46, +4.80]	500
oracle_log	CORE_SUITE	-7.814 [-8.069, -7.550]	-100.0% [-103.3%, -96.6%]	+nan [+nan, +nan]	500
rl_pilot_s0	CORE_SUITE	-1.907 [-2.133, -1.676]	-24.4% [-27.3%, -21.4%]	-9.79 [-10.99, -8.61]	500
rl_pilot_s1	CORE_SUITE	-2.068 [-2.289, -1.831]	-26.5% [-29.3%, -23.4%]	-10.03 [-11.33, -8.81]	500
rl_pilot_s2	CORE_SUITE	-2.552 [-2.774, -2.314]	-32.7% [-35.5%, -29.6%]	-8.73 [-10.03, -7.43]	500
select_m1_s0	CORE_SUITE	-3.726 [-3.945, -3.499]	-47.7% [-50.5%, -44.8%]	-14.70 [-15.93, -13.62]	500
select_m1_s1	CORE_SUITE	-3.790 [-4.014, -3.560]	-48.5% [-51.4%, -45.6%]	-14.76 [-15.96, -13.68]	500
select_m1_s2	CORE_SUITE	-3.739 [-3.961, -3.520]	-47.8% [-50.7%, -45.1%]	-14.60 [-15.82, -13.45]	500

Denominators: `overlap rate (6, full horizon)` = fraction of (scene, sample) pairs with any box overlap over the full 80-step generated horizon, scenes with initial overlap excluded, NaN for logged futures; `overlap jointly-valid` = same test restricted to sub-steps where both targets have valid logged labels (defined for generated and logged trajectories on identical intervals; the oracle_log row is the reference rate); `offroad frac` = vehicle targets only, full horizon (logged: valid steps only); cp/cv are deterministic single trajectories; learned models use 6 joint samples (32 for the audit columns).

3a. Per-seed development comparison

Development comparison (500 dev scenes; bootstrap unit = heuristic source-segment group, 95% CI)

Sampling: learned models 6 samples (joint metrics) + 32-sample audit; cp/cv deterministic (1 sample, listed separately).

Per-seed results (joint minADE@6 8 s, m; overlap and offroad rates as fractions)

method	seed	clean jminADE	clean jminFDE	core-suite jminADE	held-out jminADE	clean overlap
cp	det.	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.742 [25.024, 28.612]	26.742 [25.024, 28.612]	0.0000 [0.0000, 0.0000]
cv	det.	10.822 [10.236, 11.404]	27.732 [26.190, 29.256]	11.250 [10.708, 11.804]	13.116 [12.576, 13.676]	0.1663 [0.1345, 0.2000]
oracle_log	det.	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	—
m0_pilot_s0	0	3.776 [3.522, 4.037]	9.195 [8.468, 9.916]	7.814 [7.550, 8.069]	11.286 [10.949, 11.620]	0.1456 [0.1274, 0.1643]
m0_pilot_s1	1	3.666 [3.417, 3.915]	8.672 [8.003, 9.338]	7.308 [7.047, 7.572]	11.197 [10.827, 11.578]	0.1520 [0.1310, 0.1730]
m0_pilot_s2	2	3.765 [3.491, 4.027]	9.125 [8.428, 9.842]	7.441 [7.175, 7.698]	12.110 [11.738, 12.464]	0.1536 [0.1333, 0.1757]
mind_pilot_s0	0	3.871 [3.598, 4.137]	9.522 [8.764, 10.272]	8.095 [7.836, 8.349]	11.962 [11.655, 12.267]	0.1770 [0.1557, 0.1995]
mind_pilot_s1	1	4.020 [3.755, 4.277]	9.724 [9.001, 10.440]	8.058 [7.778, 8.329]	12.189 [11.811, 12.556]	0.1837 [0.1617, 0.2063]
mind_pilot_s2	2	3.796 [3.544, 4.053]	9.109 [8.437, 9.796]	7.455 [7.178, 7.733]	11.648 [11.278, 11.999]	0.2027 [0.1804, 0.2256]
m1_pilot_s0	0	3.561 [3.310, 3.811]	8.418 [7.739, 9.089]	4.329 [4.093, 4.565]	6.351 [6.121, 6.598]	0.1583 [0.1379, 0.1813]
m1_pilot_s1	1	3.481 [3.234, 3.724]	8.424 [7.734, 9.092]	4.235 [4.012, 4.459]	6.256 [6.011, 6.497]	0.1423 [0.1218, 0.1630]
m1_pilot_s2	2	3.495 [3.246, 3.728]	8.360 [7.680, 9.019]	4.218 [3.988, 4.447]	6.237 [6.012, 6.489]	0.1473 [0.1263, 0.1700]
select_m1_s0	0	3.331 [3.081, 3.586]	7.818 [7.131, 8.509]	4.088 [3.855, 4.321]	5.854 [5.623, 6.087]	0.0107 [0.0034, 0.0194]
select_m1_s1	1	3.273 [3.040, 3.509]	7.899 [7.269, 8.549]	4.024 [3.797, 4.246]	5.897 [5.648, 6.159]	0.0057 [0.0013, 0.0114]
select_m1_s2	2	3.334 [3.102,	8.073 [7.412,	4.076 [3.831,	5.793 [5.556,	0.0184 [0.0084,

method	seed	clean jminADE	clean jminFDE	core-suite jminADE	held-out jminADE	clean overlap
		3.564]	8.728]	4.309]	6.043]	0.0301]
extra_sft_pilot_s0	0	3.454 [3.200, 3.706]	8.442 [7.750, 9.133]	4.302 [4.062, 4.532]	6.369 [6.122, 6.608]	0.1583 [0.1379, 0.1810]
extra_sft_pilot_s1	1	3.470 [3.241, 3.691]	8.401 [7.767, 9.049]	4.207 [3.981, 4.433]	6.231 [5.994, 6.476]	0.1356 [0.1155, 0.1570]
extra_sft_pilot_s2	2	3.493 [3.238, 3.732]	8.375 [7.672, 9.053]	4.250 [4.021, 4.482]	6.174 [5.941, 6.416]	0.1436 [0.1233, 0.1653]
filtered_sft_pilot_s0	0	3.609 [3.334, 3.877]	8.667 [7.953, 9.391]	4.379 [4.150, 4.611]	6.493 [6.248, 6.752]	0.1523 [0.1323, 0.1750]
filtered_sft_pilot_s1	1	3.559 [3.321, 3.803]	8.677 [8.000, 9.368]	4.389 [4.161, 4.616]	6.501 [6.245, 6.771]	0.1430 [0.1214, 0.1653]
filtered_sft_pilot_s2	2	3.488 [3.234, 3.728]	8.411 [7.690, 9.084]	4.303 [4.070, 4.542]	6.313 [6.056, 6.580]	0.1373 [0.1172, 0.1579]
rl_pilot_s0	0	4.504 [4.225, 4.755]	11.114 [10.349, 11.874]	5.907 [5.630, 6.179]	8.360 [8.033, 8.684]	0.0528 [0.0406, 0.0647]
rl_pilot_s1	1	4.409 [4.147, 4.663]	10.856 [10.135, 11.545]	5.746 [5.477, 6.007]	8.089 [7.762, 8.404]	0.0428 [0.0315, 0.0552]
rl_pilot_s2	2	4.139 [3.864, 4.404]	10.119 [9.368, 10.894]	5.262 [4.997, 5.513]	7.559 [7.258, 7.858]	0.0625 [0.0487, 0.0773]

Q1 Joint vs independent decoder (m0 − mind; negative = joint better)

seed	clean jminADE Δ	clean jminFDE Δ	clean overlap Δ (pp)	clean overlap32 Δ (pp)
0	-0.095 [-0.289, +0.104] (-2.5%)	-0.327 [-0.900, +0.276] (-3.4%)	-3.14 [-5.32, -1.07]	-2.82 [-4.36, -1.16]
1	-0.354 [-0.548, -0.165] (-8.8%)	-1.052 [-1.599, -0.498] (-10.8%)	-3.17 [-5.33, -0.97]	-2.46 [-4.00, -0.87]
2	-0.031 [-0.232, +0.159] (-0.8%)	+0.015 [-0.614, +0.603] (+0.2%)	-4.91 [-7.08, -2.73]	-3.69 [-5.30, -1.98]

seed	clean jminADE Δ	clean jminFDE Δ	clean overlap Δ (pp)	clean overlap32 Δ (pp)
sign agreement	consistent	MIXED	consistent	consistent

Q2 Corruption-trained vs clean-trained under degraded inputs (m1 − m0; negative = m1 better)

seed	core-suite jminADE Δ	held-out jminADE Δ	core-suite jminFDE Δ	core-suite overlap Δ (pp)
0	-3.485 [-3.698, -3.270] (-44.6%)	-4.935 [-5.239, -4.643] (-43.7%)	-7.454 [-8.028, -6.901] (-43.1%)	+0.63 [-0.66, +1.94]
1	-3.073 [-3.278, -2.870] (-42.1%)	-4.942 [-5.290, -4.591] (-44.1%)	-6.655 [-7.208, -6.104] (-40.8%)	+0.92 [-0.47, +2.26]
2	-3.222 [-3.433, -3.007] (-43.3%)	-5.873 [-6.193, -5.535] (-48.5%)	-7.182 [-7.740, -6.647] (-42.7%)	-0.81 [-2.08, +0.54]
sign agreement	consistent	consistent	consistent	MIXED

Q3 Clean accuracy and physical consistency (m1 − m0 on clean)

seed	clean jminADE Δ	clean overlap Δ (pp)	clean offroad Δ (pp)	clean implausible Δ (pp)
0	-0.214 [-0.391, -0.036] (-5.7%)	+1.27 [-0.53, +3.21]	-0.14 [-0.77, +0.47]	-0.05 [-0.13, +0.00]
1	-0.185 [-0.354, -0.016] (-5.1%)	-0.97 [-3.02, +1.07]	-1.15 [-1.83, -0.52]	+0.02 [-0.03, +0.08]
2	-0.270 [-0.471, -0.067] (-7.2%)	-0.63 [-2.61, +1.48]	-2.03 [-2.66, -1.43]	+0.05 [+0.00, +0.13]
sign agreement	consistent	MIXED	consistent	MIXED

Q5a Bounded RL vs its M1 initialisation (rl − m1)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
0	+0.943 [+0.781, +1.114] (+26.5%)	+1.578 [+1.401, +1.750] (+36.5%)	-10.55 [-12.45, -8.85]	-10.42 [-11.99, -9.03]	-10.38 [-11.94, -8.99]	-1.18 [-1.69 -0.70
1	+0.928 [+0.760, +1.098] (+26.7%)	+1.512 [+1.335, +1.680] (+35.7%)	-9.95 [-11.91, -8.17]	-11.90 [-13.57, -10.34]	-11.82 [-13.53, -10.29]	-1.33 [-1.7 -0.90
2	+0.644 [+0.463, +0.831] (+18.4%)	+1.044 [+0.893, +1.197] (+24.7%)	-8.48 [-10.34, -6.68]	-9.01 [-10.40, -7.62]	-8.94 [-10.34, -7.56]	-1.16 [-1.59 -0.74
sign agreement	consistent	consistent	consistent	consistent	consistent	consi

Q5b Extra-SFT control vs M1 (extra_sft − m1)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	clea offro Δ (p
0	-0.107 [-0.244, +0.030] (-3.0%)	-0.027 [-0.087, +0.040] (-0.6%)	+0.00 [-1.57, +1.57]	+1.05 [+0.34, +1.78]	+1.06 [+0.36, +1.78]	+0.09 [-0.21 +0.40
1	-0.011 [-0.140, +0.130] (-0.3%)	-0.028 [-0.090, +0.035] (-0.7%)	-0.67 [-2.04, +0.77]	-1.03 [-1.76, -0.32]	-1.01 [-1.74, -0.29]	-0.21 [-0.56 +0.12
2	-0.003 [-0.126, +0.120] (-0.1%)	+0.032 [-0.034, +0.103] (+0.8%)	-0.37 [-1.77, +1.07]	-0.06 [-0.83, +0.67]	-0.01 [-0.77, +0.71]	+0.22 [-0.11 +0.55
sign agreement	consistent	MIXED	MIXED	MIXED	MIXED	MIXI

Q5c Reward-filtered-SFT control vs M1 (filtered_sft − m1)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	clea offro Δ (p
0	+0.047 [-0.080, +0.173] (+1.3%)	+0.050 [-0.020, +0.115] (+1.1%)	-0.60 [-2.13, +1.00]	+0.33 [-0.42, +1.11]	+0.29 [-0.44, +1.05]	+0.16 [-0.15 +0.46
1	+0.078 [-0.037, +0.192] (+2.2%)	+0.154 [+0.087, +0.224] (+3.6%)	+0.07 [-1.44, +1.50]	-1.88 [-2.68, -1.08]	-1.92 [-2.72, -1.12]	-0.10 [-0.43 +0.23
2	-0.007 [-0.126, +0.111] (-0.2%)	+0.084 [+0.007, +0.161] (+2.0%)	-1.00 [-2.61, +0.57]	-1.33 [-2.14, -0.58]	-1.29 [-2.10, -0.53]	+0.11 [-0.21 +0.44
sign agreement	MIXED	consistent	MIXED	MIXED	MIXED	MIXI

Q5d Reward-based candidate selection vs M1 (select − m1)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
0	-0.230 [-0.348, -0.108] (-6.5%)	-0.241 [-0.319, -0.164] (-5.6%)	-14.76 [-16.93, -12.86]	+0.19 [-0.43, +0.79]	+0.15 [-0.47, +0.74]	-3.26 [-3.76 -2.78
1	-0.208 [-0.322, -0.090] (-6.0%)	-0.210 [-0.283, -0.141] (-5.0%)	-13.66 [-15.67, -11.68]	+0.06 [-0.61, +0.72]	+0.04 [-0.64, +0.68]	-3.07 [-3.56 -2.57
2	-0.162 [-0.284, -0.039] (-4.6%)	-0.143 [-0.224, -0.063] (-3.4%)	-12.89 [-14.96, -11.00]	-0.59 [-1.23, +0.04]	-0.56 [-1.20, +0.07]	-3.05 [-3.54 -2.58
sign agreement	consistent	consistent	consistent	MIXED	MIXED	consi

Q5e RL vs extra-SFT control (rl − extra_sft)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
0	+1.050 [+0.864, +1.234] (+30.4%)	+1.606 [+1.431, +1.781] (+37.3%)	-10.55 [-12.63, -8.73]	-11.47 [-13.13, -9.96]	-11.45 [-13.12, -9.92]	-1.28 [-1.79 -0.80
1	+0.940 [+0.769, +1.113] (+27.1%)	+1.540 [+1.365, +1.710] (+36.6%)	-9.29 [-11.19, -7.56]	-10.87 [-12.48, -9.41]	-10.80 [-12.39, -9.35]	-1.12 [-1.52 -0.70
2	+0.647 [+0.465, +0.838] (+18.5%)	+1.012 [+0.863, +1.157] (+23.8%)	-8.12 [-9.87, -6.31]	-8.94 [-10.24, -7.58]	-8.93 [-10.23, -7.57]	-1.38 [-1.84 -0.94
sign agreement	consistent	consistent	consistent	consistent	consistent	consi

Q5f RL vs reward-filtered-SFT control (rl − filtered_sft)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
0	+0.896 [+0.715, +1.069] (+24.8%)	+1.529 [+1.369, +1.695] (+34.9%)	-9.95 [-12.00, -8.15]	-10.75 [-12.31, -9.38]	-10.67 [-12.21, -9.31]	-1.35 [-1.81 -0.89
1	+0.851 [+0.688, +1.016] (+23.9%)	+1.357 [+1.206, +1.506] (+30.9%)	-10.02 [-12.01, -8.32]	-10.02 [-11.48, -8.66]	-9.90 [-11.34, -8.53]	-1.23 [-1.66 -0.80
2	+0.651 [+0.483, +0.832] (+18.7%)	+0.959 [+0.820, +1.099] (+22.3%)	-7.48 [-9.17, -5.88]	-7.67 [-8.90, -6.43]	-7.65 [-8.87, -6.42]	-1.27 [-1.78 -0.80
sign agreement	consistent	consistent	consistent	consistent	consistent	consi

Q5g RL vs reward-based candidate selection (rl − select_m1; select draws 32 candidates per scene)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
0	+1.173 [+0.998, +1.339] (+35.2%)	+1.819 [+1.653, +1.987] (+44.5%)	+4.21 [+3.01, +5.41]	-10.61 [-12.18, -9.20]	-10.53 [-12.10, -9.13]	+2.08 [+1.6 +2.57
1	+1.137 [+0.952, +1.325] (+34.7%)	+1.722 [+1.558, +1.885] (+42.8%)	+3.71 [+2.61, +4.88]	-11.96 [-13.59, -10.42]	-11.85 [-13.51, -10.26]	+1.73 [+1.3 +2.15
2	+0.806 [+0.627, +0.995] (+24.2%)	+1.187 [+1.047, +1.332] (+29.1%)	+4.41 [+3.21, +5.81]	-8.41 [-9.75, -7.08]	-8.38 [-9.72, -7.07]	+1.88 [+1.4 +2.34
sign agreement	consistent	consistent	consistent	consistent	consistent	consi

Q6a Excessive-stopping diagnostic (descriptive; clean dev; 500 scenes; identical scenes/sampling; eligible = agent with valid logged final step and logged final displacement > 5 m; stopped = predicted final displacement < 1 m; counted per (agent, joint sample); 6 samples/scene (select_m1: best 6 of 32 by reward); torch seed 0; descriptive only, no threshold)

method	eligible (agent, sample) pairs	stopped-while-logged-moved	frequency	seeds
m1_pilot	17082	73	0.43 %	0,1,2
select_m1	17082	75	0.44 %	0,1,2
extra_sft_pilot	17082	74	0.43 %	0,1,2
filtered_sft_pilot	17082	99	0.58 %	0,1,2
rl_pilot	17082	596	3.49 %	0,1,2
m0_pilot	17082	89	0.52 %	0,1,2

Q6b Post-training behaviour summary (clean dev; scene-level means over seeds; 6-sample metrics unless 32 noted)

method	speed_mean (m/s)	GT speed_mean	final_disp_mean (m)	GT final_disp	final / G
m1_pilot	6.61	6.66	51.08	51.56	0.991
select_m1	6.52	6.66	50.50	51.56	0.979
extra_sft_pilot	6.57	6.66	50.78	51.56	0.985
filtered_sft_pilot	6.56	6.66	50.66	51.56	0.982
rl_pilot	5.78	6.66	44.97	51.56	0.872

Sampling cost and wall-clock (clean condition)

Wall-clock seconds per scene are DESCRIPTIVE measurements logged by evaluate.py on a GPU shared with unrelated jobs; they do not establish a reliable relative speed ranking. The deterministic cost driver is the number of joint samples drawn per scene (and, for select_m1, the reward scoring of all 32 candidates).

method	samples drawn per scene	s / scene (descriptive)
cp	1	0.0000
cv	1	0.0000
extra_sft_pilot_s0	6	0.0073
extra_sft_pilot_s1	6	0.0072
extra_sft_pilot_s2	6	0.0074
filtered_sft_pilot_s0	6	0.0074
filtered_sft_pilot_s1	6	0.0074
filtered_sft_pilot_s2	6	0.0075
m0_pilot_s0	6	0.0038
m0_pilot_s1	6	0.0032
m0_pilot_s2	6	0.0032
m1_pilot_s0	6	0.0032
m1_pilot_s1	6	0.0032
m1_pilot_s2	6	0.0032
mind_pilot_s0	6	0.0031
mind_pilot_s1	6	0.0031
mind_pilot_s2	6	0.0032
oracle_log	1	0.0000
rl_pilot_s0	6	0.0078
rl_pilot_s1	6	0.0075
rl_pilot_s2	6	0.0075
select_m1_s0	32 + reward scoring of 32	0.0521

method	samples drawn per scene	s / scene (descriptive)
select_m1_s1	32 + reward scoring of 32	0.0512
select_m1_s2	32 + reward scoring of 32	0.0512

Per-condition joint minADE@6 (mean over seeds; per-seed values in summary.json)

condition	cv	m0_pilot	mind_pilot	m1_pilot	select_m1	rl_pilot
clean	10.822	3.736 (±0.049, n=3)	3.896 (±0.093, n=3)	3.513 (±0.035, n=3)	3.313 (±0.028, n=3)	4.351 (±0.155, n=3)
missing_0.1	10.814	3.892 (±0.076, n=3)	4.096 (±0.057, n=3)	3.486 (±0.033, n=3)	3.357 (±0.048, n=3)	4.370 (±0.125, n=3)
missing_0.3	10.768	4.256 (±0.105, n=3)	4.584 (±0.149, n=3)	3.540 (±0.039, n=3)	3.475 (±0.054, n=3)	4.378 (±0.113, n=3)
missing_0.5	10.762	4.781 (±0.125, n=3)	5.178 (±0.223, n=3)	3.677 (±0.063, n=3)	3.549 (±0.027, n=3)	4.707 (±0.247, n=3)
noise_0.1	10.941	6.001 (±0.216, n=3)	6.121 (±0.441, n=3)	4.060 (±0.109, n=3)	3.942 (±0.102, n=3)	5.424 (±0.265, n=3)
noise_0.3	11.631	11.647 (±0.562, n=3)	12.100 (±0.766, n=3)	4.876 (±0.046, n=3)	4.615 (±0.048, n=3)	6.842 (±0.395, n=3)
noise_0.5	12.586	14.550 (±0.679, n=3)	15.139 (±0.435, n=3)	5.925 (±0.173, n=3)	5.439 (±0.128, n=3)	8.110 (±0.510, n=3)
missing_0.7	11.427	5.955 (±0.181, n=3)	6.506 (±0.404, n=3)	4.207 (±0.010, n=3)	4.021 (±0.081, n=3)	5.410 (±0.259, n=3)
noise_1	16.053	18.395 (±1.357, n=3)	18.797 (±0.496, n=3)	9.349 (±0.115, n=3)	8.450 (±0.098, n=3)	11.273 (±0.413, n=3)
compound_0	11.868	10.243 (±0.145, n=3)	10.496 (±0.626, n=3)	5.288 (±0.033, n=3)	5.074 (±0.051, n=3)	7.326 (±0.406, n=3)

Preregistered thresholds (m1 vs m0, per seed)

seed	core-suite rel. Δ (need ≤ −5%)	clean rel. Δ (need ≤ +3%)	core-suite overlap Δ pp (need ≤ +0.5)	verdict
0	-44.6% [-47.3, -41.9]	-5.7% [-10.3, -0.9]	+0.63 [-0.66, +1.94]	FAIL
1	-42.1% [-44.8, -39.3]	-5.1% [-9.7, -0.4]	+0.92 [-0.47, +2.26]	FAIL

seed	core-suite rel. Δ (need ≤ −5%)	clean rel. Δ (need ≤ +3%)	core-suite overlap Δ pp (need ≤ +0.5)	verdict
2	-43.3% [-46.1, -40.4]	-7.2% [-12.5, -1.8]	-0.81 [-2.08, +0.54]	PASS

8. Limitations

Limitations (standing; updated as stages complete)

- **No real-data results yet.** Every number produced so far comes from SYNTHETIC fixtures used to verify the implementation; they are labelled as such and are not scientific results.
- **Simplified reproduction.** Compact model (1.8 M params vs MotionLM's 9 M), far fewer training steps, a subset of WOMD shards, no NMS/k-means mode aggregation, six raw samples for local metrics. Not an official challenge submission; the official metrics package could not be installed here (DECISIONS D-019a), so the custom metrics are audited on analytic cases and oracles only.
- **Benchmark-provided targets.** The two targets are the dataset's `objects_of_interest` pair; this is evaluation metadata, not an onboard target-selection algorithm.
- **Open-loop context.** Context agents are not simulated; only the two targets interact (2 Hz joint autoregression). Interpolating 2 Hz outputs to 10 Hz for geometric checks does not make the model a reactive 10 Hz policy.
- **Synthetic corruption.** Missing frames and Gaussian position noise are stress scenarios, not calibrated Waymo sensor-error distributions; they say nothing about raw camera/LiDAR perception.
- **Offroad metric.** Road-edge polylines with a sign convention calibrated by the log-playback oracle; vehicles only; centre-point test (no corner test); invalid maps excluded and counted.
- **Inference unit.** Scenario-level paired bootstrap; WOMD does not expose the originating run segment, so independence between windows is assumed, not verified.
- **Seeds.** Single-seed evidence unless the pilot budget supports three seeds (planned: 3).
- **RL scope.** REINFORCE with LOO baseline/KL/replay over two agents and a 16-step episode; rewards are deterministic generated-trajectory diagnostics. Differs from the all-agent closed-loop setting of the Waymo RL paper.
- **Source-segment overlap.** Identified only through SDC motion n-gram matching (D-025); train scenes sharing a segment with dev were removed, final shares none, but same-segment windows remain inside each split, so the scenario-level bootstrap slightly overstates independence.
- **Overlap-metric reference rate.** On jointly valid intervals the 2 Hz-waypoint OBB test flags 1.0 % of logged joint futures on dev (D-028a: 3 annotation-level vehicle–pedestrian contacts, 2 heading-inference artefacts). The full-horizon rate on logged futures is undefined (invalid labels carry no geometry); model `overlap_rate` (full horizon) and `overlap_rate_jointvalid` (identical intervals as the logged reference) are both reported with their denominators.

- **Mixture training as regulariser.** The pilot over-fits after ≈ 30 epochs (D-027); corruption mixture training also augments the data, so part of any M1 gain may be regularisation rather than corruption awareness. The matched extra-SFT control and clean guardrail bound, but do not remove, this confound.
- **Overlap reference rate is not a lower bound (D-028a).** It is the same approximate test applied to logs; it mixes annotation contacts and heading-inference artefacts, and vehicle–pedestrian pairs dominate it. The RL overlap reward is therefore an approximate geometric-consistency objective, not a verified safety measure.
- **Bootstrap unit.** CIs resample heuristic source-segment groups (D-025a); the grouping is derived from SDC motion matching and cannot guarantee independence.
- **RL outcome (D-031).** Bounded REINFORCE post-training lowered box overlap but worsened clean accuracy by 18–27 % and diversity, with ≈ 8× more excessive stopping; by the frozen rule the RL effect does not count, and the inference-time candidate-selection control dominates it. Reward curves during training are not held-out improvements.
- **Timings.** Seconds-per-scene figures were measured on a GPU shared with unrelated jobs; they are descriptive and do not rank methods by speed. Sampling counts per scene are the cost driver.